Is the Rent Too High? Land Ownership and Monopoly Power*

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Abstract

Pricing power in real estate markets can reduce housing supply and redevelopment relative to the social optimum. We show how popular redevelopment subsidies and zoning regulations interact with pricing power. Using building-level rental income data from NYC, we find that increasing concentration is correlated with increasing prices. Finally, we use the model to estimate the first building-level housing elasticity, finding that markups account for between ten and thirty percent of rents in the city.

Keywords: monopolistic competition, market power, concentration, housing demand, redevelopment subsidies, zoning

JEL Classification: R31, R38, L13

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1 Introduction

Property rights grant landowners exclusive use over parcels of land. Since Adam Smith, economists have considered whether this arrangement endows land owners with monopoly pricing powers. If parcels of land or the houses built on those parcels are perfectly substitutable and a large number of parcels or houses are made available by many individual landowners in friction-free markets, then land and housing markets might be best approximated by perfect competition. Alternatively, differentiation, frictions, and concentration may make land and housing markets less than perfectly competitive. While the recent rise of institutional landlords has increased interest in the effects of concentration and pricing power, the theoretical and empirical implications of pricing power in real estate are poorly understood.

This paper investigates the economic impact of pricing power due to land rights. We answer two questions: how should pricing power alter our understanding of urban land use policies and is this power economically meaningful? We show how pricing power, derived from the combination of exclusive land ownership rights and idiosyncratic renter tastes, interacts with redevelopment incentives and zoning regulations and examine the possibility that restrictions on land ownership concentration can reduce rents. Using data on multi-unit residential rental buildings in New York City (NYC), we estimate a positive concentration-rent correlation. Estimating the model on our data, we measure finite building own-price elasticities and markups to be between one third and one tenth of rents in our data.

We model the construction and rental of housing units in the presence of idiosyncratic preferences for locations, which generates downward-sloping demand for units at individual parcels or buildings, which we call buildings' *residual demand*. While aggregate demand can be downward sloping in competitive frameworks, when *residual* demand is downward sloping, landlords with exclusive right to rent at those plots have pricing power, even when profits from rents are capitalized into land values. This restricts supply on three margins: reducing the size of new buildings, removing existing units from the market when demand is low, and reducing the pace of redevelopment when demand is high, even when such redevelopment is socially optimal given costs.[1] We call this third margin 'redevelopment failure.'[2] In Appendix A.8, we show how markups and these

[1]Vogell (2022) documents the second margin in interviews with New York City landlords.

[2]This is related to the 'lock-in' effect of vintage capital in Boustan, Bunten, and Hearey (2018). Siodla (2015); Hornbeck and Keniston (2017) discuss redevelopment in the presence of natural disasters. A large

margins of supply constraints act as a dispersing force in a monocentric city.

We explore the implications on three types of urban policies: redevelopment subsidies, ownership concentration, and zoning. First, because of redevelopment failure, subsidizing the decision to redevelop can improve welfare. We explore a low-information alternative to optimal subsidies that does not require knowledge of demand and supply elasticities: pairing lump-sump development subsidies with promised rent reductions. These policies echo the common pairing of local tax subsidies with inclusionary zoning. While such policies are often viewed as equity-oriented, we offer a potential efficiency justification.

Second, we explore restrictions on ownership concentration, as recently proposed in Berlin (Stone, 2019). With non-decreasing marginal cost, concentrated owners raise markups, internalizing the impact of one parcel's pricing decision on profits at other parcels.[3] We extend Nocke and Schutz (2018b) to find conditions under which increased concentration also generates increases in prices for all other products—in our case parcels. This provides a theoretical basis for work studying the impact of landlord and developer concentration (Cosman and Quintero, 2021; Raymond et al., 2016).

Third, markups amplify the effect of zoning, as zoning constraints at one parcel raise rents at other unzoned parcels through markups. At the same time, pricing power attenuates the impact of up-zoning through redevelopment failure, and because redeveloped upzoned parcels revert to monopolistic rather than efficient levels.

To quantify the potential forces in our model, we construct a building-level dataset for multi-unit residential buildings in NYC using a combination of public and scraped data. While Manhattan may be a unique environment, outer borough density is similar to other urban markets and provides similar results. First, we find *non-causal* correlations in the data consistent with the predictions of our model: a 10% increase in Census tract concentration is correlated with a 0.4-0.6% increase in average building rents, even accounting for time-invariant building characteristics.

We then estimate the structural parameters of our model to calculate building-level own-price elasticities (OPEs) of demand—not previously estimated in the literature—to evaluate the quantitative scope of markups.[4] We estimate OPEs using three sets of instruments with distinct sources of variation and identifying assumptions, as in Hornbeck and

previous literature (e.g., Rosenthal and Helsley, 1994) explores redevelopment under perfect competition.

[3]This result builds on Nocke and Schutz (2018a), part of a growing literature on multi-product oligopoly (Affeldt, Filistrucchi, and Klein, 2013; Jaffe and Weyl, 2013; Nocke and Schutz, 2018a)

[4]Our estimation method is based on differentiated product demand estimation (e.g., Berry, 1994). See Kuminoff, Smith, and Timmins (2013) for a literature overview.

Moretti (2018). As is common in estimates of housing demand, we use rival characteristics ('BLP instruments') that affect the landlord's ability to markup rent. We also use a static version of the synthetic tax instrument in Watson and Ziv (2025) and also historic tax and construction costs as alternative cost shifters. The complementary strategies yield similar parameter estimates, with median OPEs between $[-9, -3]$.[5]

We apply these results to characterize markups. Housing presents a unique setting for markup estimation. Buildings' sizes are constrained by policy (zoning) and, because housing is durable, determined long before current renters demand housing.These factors complicate traditional estimates of markups. Importantly, our model gives guidance for how OPEs can be estimated in the presence of these policy and capacity constraints. We find a median markup over marginal costs—including amortized purchase, maintenance, leasing costs and outside options—between one-tenth and one-third of the rent.

A large literature explores the impact of housing supply constraints Hsieh and Moretti (2019); Glaeser (2007); Glaeser and Gottlieb (2008). This paper introduces pricing power as an additional source of housing constraints. While Proposition 2 shows that markups can amplify policy constraints in general equilibrium, pricing power also reduces supply in the absence of policy constraints and accounts for a 10-30% wedge between prices and costs. Appendix A.8 shows that pricing power acts as a dispersing force, flattening and widening the city. We find markups for new, zoning unconstrained buildings approximate those implied at older, zoning constrained buildings, suggesting that the latter group may not be far from monopoly-optimal price, and that reductions in policy constraints alone will not eliminate price-cost wedges.

Recent empirical work has focused on housing ownership concentration (Cosman and Quintero, 2021; Raymond et al., 2016). In addition to evidence in this vein, we provide a theoretical basis for this nascent work, establishing evidence of markups in housing and both a rationale for how and necessary conditions for when concentration impacts prices.

Our paper also makes a contribution to the empirical literature on housing demand elasticities. Previous estimates of housing demand have focused on the elasticity of *aggregate* demand—i.e. the housing-consumption trade-off. Our object of interest is the price elasticity of demand of a given building—that is, in consumers' willingness to substitute *between* rival buildings. These are to our knowledge the literature's first estimates of building own-price elasticities. Our estimates also imply aggregate demand

[5]In previous versions of this draft, we estimated more complex nested and mixed logit models with similar results.

elasticities that are consistent with those previous estimates.[6]

Section 2 lays out our model and Section 3 uses the model to derive results on housing policy. Section 4 introduces our data. Section 5 estimates the concentration-rent relationship in our data. Section 6 estimates own-price elasticities and Section 7 concludes.

2 Model

We first set up the optimization routines for each agent in our model. A social planner sets a set of construction subsidy rules for each parcel. Next, developers are endowed with spatially differentiated parcels and decide whether to (re)develop before selling.[7] After developers make decisions, landlords buy buildings and post rents for their building. Renters endowed with income and idiosyncratic tastes for the differentiated parcels see posted prices and choose residence locations. This setup allows us to consider how redevelopment responds to long-run changes in demand and explore how pricing, leasing, and redevelopment decisions are impacted by pricing power. We abstract from dynamic rent flow considerations, but our a model is consistent with a dynamic model with constant capital prices and no depreciation.[8]

Section 2.2 characterizes the subgame-perfect equilibrium of the model. Appendix A.8 explores extensions, including a competitive benchmark case, a version the following model embedded into a monocentric city, and the introduction of entry, exit, and risk.

2.1 Setup

Parcels, Developers, and Landlords The space, a city, is comprised of a set discrete parcels, $\mathcal{A} = \{a_0, a_1, a_2, \ldots, a_J\}$, that differ according to their underlying quality a, drawn without replacement from a distribution $G(a)$. Higher values of a have higher amenity value to renters. We refer to a as "location quality" and differences in a as vertical differentiation in parcels. A location's realized quality a will also be used henceforth to index each location in the set $\mathcal{A}$. In the exposition that follows, we make the simplifying assumption that a is exogenous.[9] We set a_0 as living out of the city (i.e., an outside-option).

[6]Following Berry and Jia (2010), we estimate that if *all* building rents increased by 1%, aggregate rental demand responds with an elasticity between $[0.60, 2.4]$ (Albouy and Ehrlich, 2018). Using hedonic approaches, Gyourko and Voith (2000) and Chen, Clapp, and Tirtiroglu (2011) find elasticities above 1.

[7]Parcels may be "greenfield" with zero units or "brownfield" with some existing structure.

[8]See equation 2.5 in Rosenthal and Helsley (1994).

[9]In our empirical work we consider that in the data building and parcel characteristics are a mix of endogenously chosen and exogenously given. In Appendix A.8.6 we allow a to be endogenously chosen.

Each parcel may have a building with an initial number of units, $q_{a,0} \geq 0$, and is owned by a unique developer, indexed by $\mathsf{d} \in \mathsf{D}$, who chooses whether to sell the parcel 'as-is,' in which case the number of units remains $q_{a,0}$, or redevelop to $q_{a,1}$ before selling to a landlord who leases the units.[10] Each developer—subsequently each landlord—owns one parcel.

Redevelopment requires a cost $C_a^{\mathsf{d}}(q_{a,1})$ that is an a-dependent step function, including a fixed cost $C_a^{\mathsf{d}}(0) \geq 0$, a constant marginal cost below the zoning limit denoted $c_a^{\mathsf{d}} > 0$ and a higher cost above the limit $c_a^{\mathsf{d}} + \theta_z$, where θ_z is an arbitrarily high cost of exceeding the zoning regulations.[11] Developers choose whether to redevelop and conditional on redevelopment, quantity $q_{a,1}$ to maximize profit:

$$\pi_a^{\mathsf{d}} = \max_{\mathbb{1}_{redev}, q_{a,1}} \begin{cases} s_a(a, q_{a,0}) & \text{if } \mathbb{1}_{redev} = 0 \\ s_a(a, q_{a,1}) - C_a^{\mathsf{d}}(q_{a,1}) + \mathsf{S}_a & \text{if } \mathbb{1}_{redev} = 1 \end{cases} \tag{1}$$

where $\mathbb{1}_{redev}$ is an indicator function for the choice to redevelop, s_a is the realized sale price of the building, which would be $s_a(a, q_{a,0})$ if sold 'as-is' or $s_a(a, q_{a,1})$ if sold after redevelopment to size $q_{a,1}$, and S_a is a parcel-specific redevelopment subsidy.

For each parcel, a competitive fringe of potential landlords, indexed by $\mathsf{f} \in \mathsf{F}$, bid for sole the right to lease units and set rents in the building at at that location. Landlords who purchase buildings provide a mass of renters housing at a positive step-function cost $C_a^{\mathsf{f}}(q_a^{\mathsf{f}})$ with positive, constant marginal cost c_a^{f} below the developer-prescribed capacity and $c_a^{\mathsf{f}}+\theta_k$ above, where q_a^{f} is the mass of renters the landlord accommodates in equilibrium. θ_k is presumed to be arbitrarily higher than the above-zoning marginal cost of construction. Landlord f's profit from parcel a is:

$$\pi_a^{\mathsf{f}} = r_a \cdot q_a^{\mathsf{f}} - C_a^{\mathsf{f}}(q_a^{\mathsf{f}}) - s_a \tag{2}$$

where total revenue is rent r_a collected times the quantity offered to renters q_a^{f}, which is constrained to be at or below the quantity chosen by the developer, $q_{a,\mathsf{d}}$.[12]

By modeling short and long run (zoning) capacity as arbitrarily high marginal costs beyond a determined level, following Maggi (1996), we recover a pure strategy equilibrium, avoiding a well-known existence problem in differentiated product markets with

[10] We refer to buildings, and parcels interchangeably throughout.

[11] We allow for this hypothetical for technical reasons discussed below.

[12] We do not allow for price discrimination. An alternative dynamic search setting where landlords price and market units individually retains our static setting's results.

strict capacity constraints (Wauthy et al., 2014).[13] Strict constraints at one building can provide incentive for an unconstrained building to deviate to off-path higher prices, which jeopardizes the existence of pure strategy equilibrium Benassy (1991).

While developer-driven rezoning is common enough to potentially justify our soft constraint approach, hard constraints might arguably be more realistic in our context. Nevertheless, these particular dis-economies of scale only govern "off the equilibrium responses which are never observed in practice" (Goldberg, 1995) and so, in equilibrium, prices are set such that there is no rationing and no unprofitable sales. Looking ahead toward our empirical results, once a pure strategy equilibrium exists, "the effects of capacity constraints can be safely ignored for estimation purposes" (Froeb, Tschantz, and Crooke, 2003)–that is, all options are available to all consumers.

Alternative rules to avoid nonexistence include assuming renters face increasing search costs per building but do not observe building capacity (Schulz, 1999) or assuming linear demand and small product shares (Yin and Ng, 1997; Schulz, 2000; Yin and Ng, 2000). Additionally, allowing for off-equilibrium arbitrage through post-lease exchanges (e.g., subleasing agreements) allows us to forgo informational assumptions in Schulz (1999). We elaborate on our current approach and discuss these alternatives in Appendix A.1.

Renters A unit mass of heterogeneous renters, indexed by $i \in \mathsf{N}$, with utility derived from consumption and location amenities. Renters draw idiosyncratic tastes for each location, $\epsilon_{i,a}$, from a standard type-one extreme value distribution.[14] Thus, utility of renter i at a is:

$$U_i(a) = F(a, r_a) + \epsilon_{i,a}. \tag{3}$$

Rent is in the utility function because we have implicitly substituted the budget constraint for consumption. Renters choose among all locations a to maximize utility taking amenities, rents, and personal income as given.

Demand Market demand is a function of the vector of rents and building qualities:

$$q_a^D = D_a(\vec{\mathbf{a}}, \vec{\mathbf{r}}) = \frac{\mathsf{e}^{F(a,r_a)}}{\sum_{\tilde{a}\in\mathcal{A}} \mathsf{e}^{F(\tilde{a},r_{\tilde{a}})}}. \tag{4}$$

[13]Closer to our setup, Greenfield and Sandford (2018) models N products, asymmetric cost and a prohibition on rationing.

[14]We follow Chamberlin (1933) in founding pricing power for individual plots in heterogeneous preferences. Several alternative micro-foundations are possible, including search and matching frictions. Adopting these isomorphic stories would not change our estimation results, although doing so would naturally change their interpretation.

We write $q_a^D = D_a(r_a)$, suppressing other arguments to ease notation. We can invert the system of demand equations to recover inverse demand functions for each building, $r_a = R_a(q_a)$, that are differentiable with respect to quantity, $\frac{\partial R_a}{\partial q_a}(q_a)$. We refer to building-level demand as the building's *residual demand*, in contrast to aggregate or market-level demand. Anticipating the price-setting decision, the equilibrium price elasticity of demand is:

$$\varepsilon_a = \frac{\partial D_a(r_a)/\partial r_a}{D_a(r_a)/r_a} = \frac{\partial F(a, r_a)}{\partial r} \cdot r_a \cdot (1 - D_a(r_a)) < 0. \tag{5}$$

The own-price elasticity of residual demand, which is derived from renters' idiosyncratic preferences for buildings, is key to pricing power and markups. Appendix A.8 explores an alternative model where this residual demand is completely elastic.

Social Welfare We define the *local* social welfare as the economic surplus at each location a, the sum of both renter (consumer) surplus and developer (producer) surplus:

$$\mathsf{W}_a(q_a^e) = \int_0^{q_a^e} \left(R_a(q^e) - c_a^{\mathsf{f}}(q^e) - \mathbb{1}_{redev} \cdot c_a^{\mathsf{d}}(q^e)\right) \mathsf{d}q, \tag{6}$$

where q_a^e is the equilibrium quantity for building a, which depends on the redevelopment decision (i.e. value of $\mathbb{1}_{redev}$). We assume every other building is at its equilibrium quantity. Note that any redevelopment subsidy, S_a, is paid for *ex post* of all decisions and drops out from the local welfare calculation. We define *total* social welfare as the total economic surplus in the city, where total social welfare is $\mathsf{W} = \sum_{a \in \mathcal{A}} \mathsf{W}_a$, save the outside option.

We consider a city planner who wishes to maximize W in order to achieve allocative efficiency in the city and can regulate and subsidize the rental market. However, the city planner does not know the preferences or opportunity costs of market participants and cannot use unobservable information when setting policies.

We define a *locally welfare improving redevelopment* to be a redeveloped structure that has higher local welfare relative to *the existing structure*; i.e., $\mathsf{W}_a(q_a^e(\mathbb{1}_{redev} = 1)) > \mathsf{W}_a(q_{a,0}^e)$. We define a *locally welfare improving policy* to be a subsidy S_a or additional policy such that $\mathsf{W}_a(q_a^e)$ is higher under the policy compared to *laissez faire*, and to be welfare improving overall if W is higher under the policy.

2.2 Equilibrium

A subgame-perfect equilibrium is defined by a schedule of redevelopment decisions and quantities, posted rents, and renter location choices such that conditional on the actions of other agents and rivals, no developer can increase profits by making a different redevelopment decision, no landlord can increase profits by changing posted rents, and no renter can increase utility by choosing to pay posted rents at any other parcel. We assume θ_z and θ_k are high enough so no post-developer or above zoning capacity is installed.

2.2.1 Joint Landlord-Developer Problem

Because it most closely tracks the canonical monopolist problem where quantities are set to maximize profit, we first consider the case of a single landlord-developer, maximizing the joint profits of the two agents. With the sale price transfer cancelling, the joint landlord-developer maximizes profits from renting net of costs of any redevelopment:

$$\pi = \max_{\mathbb{1}_{redev}, q_a} \begin{cases} r_a(q_a) \cdot q_a - C_a^{\mathsf{d}}(q_a) - C_a^{\mathsf{f}}(q_a) + \mathsf{S}_a & \text{if } \mathbb{1}_{redev} = 1 \\ r_a(q_a) \cdot q_a - C_a^{\mathsf{f}}(q_a) & \text{if } \mathbb{1}_{redev} = 0 \end{cases} \tag{7}$$

where quantity chosen is $q_a \equiv q_{a,0} \leq q_{a,0}$ if $\mathbb{1}_{redev} = 0$ and $q_a \equiv q_{a,1} \leq q_{a,z}$ if $\mathbb{1}_{redev} = 1$.

If the landlord-developer chooses not to redevelop, their maximum quantity is fixed at $q_{a,0}$, but they may choose not to rent all units. They will price on the demand curve at or below $q_{a,0}$, conscious of the fact that per-unit price is falling with quantity $\partial R_a(q_a)/\partial q_a < 0$.

Redevelopment and Redevelopment Failure Redevelopment occurs if the associated cost is less than the difference between the maximal marginal profits with and without it:

$$C_a^{\mathsf{d}}(q_{a,0}^*) < [r_a(q_{a,1}^*) \cdot q_{a,1}^* - C_a^{\mathsf{f}}(q_{a,1}^*)] - [r_a(q_{a,0}^*) \cdot q_{a,0}^* - C_a^{\mathsf{f}}(q_{a,0}^*)], \tag{8}$$

for an quantity $q_{a,1}^* \leq q_{a,z}$ that is optimal conditional on choosing redevelopment and a quantity $q_{a,0}^*$ that is optimal conditional on not redeveloping.

An immediate implication is that redevelopment only occurs when the optimal leasing quantity is higher than the original quantity: $q_{a,1}^* > q_{a,0}^*$. Furthermore, when zoning binds on the initial building, $q_{a,z} < q_{a,0}$, there is no redevelopment.

If this criterion for redevelopment is met, then the problem becomes the typical monopolist problem with composite marginal cost $c_a^{\mathsf{d}}(q_a) + c_a^{\mathsf{f}}(q_a)$, so that the landlord-developer sets q_a^* such that $\frac{\partial R_a}{\partial q_a}(q_a^*)q_a^* + r_a(q_a^*) = c_a^{\mathsf{d}}(q_a^*) + c_a^{\mathsf{f}}(q_a^*)$. If the market was perfectly competitive, then landlords would set $q_a > q_a^*$ such that $r_a(q_a) = c_a^{\mathsf{d}}(q_a) + c_a^{\mathsf{f}}(q_a)$. This leads us to the following definition of what we term local redevelopment failure.

Definition 1. *Local redevelopment failure* occurs at location a if developers at a choose not to redevelop while the redevelopment would be locally welfare improving.

Note, our definition ignores additional competitive forces that might reduce rents and increase quantities at other buildings and thus increase social welfare.

Appendix A demonstrates that for any positive, increasing, and convex cost functions $C_a^{\mathsf{d}}(q)$ and $C_a^{\mathsf{f}}(q)$, there exists a demand system for location a that results in local redevelopment failure. When costs are especially high, neither the social nor private benefits of redevelopment to $q_{a,1}^*$ outweigh the redevelopment costs. Conversely, if demand is especially high or redevelopment costs are especially low, then redevelopment can be both socially and privately optimal. However, the reduction in rents from moving from $q_{a,0}^*$ act as an additional friction that may prevent landlord-developers from pursuing redevelopment in certain intermediate cases. For example, if a neighborhood is more in demand than when the building was originally constructed, then the planner may want to see redevelopment, as the total social benefits outweigh the redevelopment costs, but landlord-developer profit is maximized by renting the building *as-is*.

Notably, this discrepancy between socially optimal and privately optimal decisions only occurs with finitely elastic (i.e., downward sloping) building-level demand.[15] That is, the landlord and developer's decisions can only be distortionary to the extent to which a building's residual demand curve is downward sloping.

Mark-ups and quantity restrictions Depending on demand, the landlord-developer sets quantities at one of three levels. First, if demand is low enough relative to $q_{a,0}$, the landlord-developer chooses not to redevelop and further restricts quantities by holding units back from the market. In this case, they set $q_{a,0}$ such that $\frac{\partial R_a}{\partial q_a}(q_a) \cdot q_a + \frac{\partial R_a}{\partial q_a}(q_a) = c_a^{\mathsf{f}}(q_a)$. Here, the equilibrium rent is marked-up over the marginal cost of leasing, and can be calculated according to the Lerner rule: $1/\varepsilon_a$. Second, in the case where demand is high enough relative to $q_{a,0}$ to merit redevelopment, landlord-developers set $q_{a,1}$ such that $\frac{\partial R_a}{\partial q_a}(q_a) \cdot q_a + \frac{\partial R_a}{\partial q_a}(q_a) = c_a^{\mathsf{f}}(q_a) + c_a^{\mathsf{d}}(q_a)$. Here, the equilibrium rent is marked-up over the composite marginal cost of leasing *and* redevelopment, and can be calculated according to the Lerner rule. Finally, if demand is high enough such that $q_{a,0}^* = q_{a,0}$ yet redevelopment is still suboptimal for the landlord-developer, then no units will be held back and the price will be $r_a(q_{a,0})$. At this corner, the Lerner rule as a measure of markups does not apply.

[15]We note that while social planners may also want to change the quantity set by the monopolist conditional on redevelopment (i.e. set $q_{a,1} > q_{a,1}^*$), we define redevelopment failure as the decision to redevelop or not, taking $q_{a,1}^*$ as given. As such, redevelopment failure never applies to buildings where the monopoly-optimal quantity is below the size of the existing structure. For these buildings, even if redevelopment did occur, developers would set $q_{a,1}^* < q_{a,0}$. For a greater theoretical discussion, see the above cited literature on rent control.

2.2.2 Separate Landlord and Developer Problems

To solve the disjoint problem, we begin with the landlord's optimal pricing and supply strategy and work backwards to the redevelopment problem. If the developer can anticipate the landlord's problem, then the approaches are equivalent as the competitive fringe of landlords bid the price to the optimal monopolistic profit amount.

Once a building is sold, a landlord's maximum supply is bounded by $q_{a,1}$. The landlord prices on the demand curve at or below the quantity $q_{a,1}$ to maximize monopoly profits according to the rule $\frac{\partial R_a}{\partial q_a}(q_{a,f}) = c_a^{\mathrm{f}}(q_{a,f})$. Because potential landlords bid competitively on each plot, the sale price of such a plot becomes

$$s_a(q_{a,1}) = r_a(q^*_{a,f}) \cdot q^*_{a,f} - C_a^{\mathrm{f}}(q^*_{a,f}) \quad \text{s.t.} \quad q^*_{a,f} \leq q_{a,1}. \tag{9}$$

Substituting the expression for $s_a(q_{a,1})$ into the landlord's profit condition (eq. 2), it becomes clear that the profit maximizing choice of $q_{a,1}$ in this case is the same as in the joint landlord-developer decision. Although the landlord makes a separate decision, units above the optimal number to rent do not increase the sale price of the building. Were the joint decision-maker to choose not to redevelop, the landlord would choose a quantity $q^*_{a,f} \leq q_{a,0}$, and the developer sees no benefit from redevelopment. Were the joint decision maker to redevelop, the chosen quantity q_a is exactly the point where the marginal increase in sale price equals the marginal increase in cost. In this case, the landlord always chooses the corner $q^*_{a,f} = q^*_{a,1}$.[16] While the landlord appears to be choosing a quantity where marginal revenue is above marginal cost, the joint decision maker case instructs us that the quantity is at the point where marginal revenue equals the combined marginal cost of both landlord and developer.

2.3 Extensions: Monocentric City, Entry, Exit, and Risk

In Appendix A.8, we discuss several extensions of our model, such as adapting a monocentric city model, introducing entry and exit, and risk. Our results of markups of rent over marginal cost survives as long as our model includes downward sloping residual demand and landlord exclusivity in renting out the units. Without entry, our model in this section serves as a reasonable medium-run model of a built urban environment with use laws.[17] The Appendix details how entry and exit from other real estate markets (e.g., condominium or commercial markets) would generate a cross-market profit equalization

[16]Note of course that redevelopment $q^*_{a,f} < q^*_{a,1}$ would imply over-development.

[17]In NYC, for example, fewer than 100 buildings in the city apply for condo conversions each year.

condition but still feature markups. This holds when we allow free entry overall and when we adapt a monocentric city framework, where entry on a particular margin, the city's fringe, pins down the landlord-developer profit gradient. This latter model also makes clear how pricing power acts as a dispersing force reducing the supply of housing, flattening and widening the city. The Appendix also discusses how risk enters this model.

2.4 Summary Discussion

The price-setting power of landlords and developers affects supply in three ways. First, in cases where demand is low, landlords withhold units from the market. Vogell (2022) documents this behavior, which is also isomorphic to setting rent at a high enough level such that landlords endure nonzero vacancy spells when lower prices would find renters sooner. Second, redevelopment may occur in fewer cases than would be the case if redevelopment decisions were made on a socially optimal basis. Third, when buildings are redeveloped, developers reduce quantities relative to an efficient benchmark in order to maximize the combined profits of redevelopment and leasing.

Developers build to the size at which marginal increase in sale value meets marginal costs, and landlords rent out the full building at market rates, while making no profits. However, the joint problem demonstrates how this behavior is equivalent to a reduction in quantity relative to an efficient baseline. While in any given period of time, few buildings are redeveloped, the model instructs that at the time of construction, those buildings were constructed in a manner which restricted supply relative to the contemporary demand.

An implication of these findings is that the Lerner Rule only maps demand elasticities to markups in cases where redevelopment has recently occurred and buildings are not at the zoning limit, or where units are withheld from the market. In the former case, markups are over the combined marginal cost of development and leasing, while in the latter case markups are only over the cost of leasing. This informs the set of buildings on which we calculate markups in Section 6.

3 Policy Implications

We assess the effects of several policies in the context of markups. First, we discuss redevelopment subsidies, and, second, the impact of zoning in the presence of markups.[18]

[18] We consider the effects of redevelopment subsidies, but we ignore distortions that finance these policies, which are beyond the scope of this paper.

3.1 Redevelopment Subsidies

Subsidies for redevelopment exist at the federal, state, and local levels. At the Federal level, the Low-Income Housing Tax Credit gives tax credit for the construction or rehabilitation of low-income housing and has been given for over 36,000 buildings accounting for nearly 2.5 million housing units (HUD, 2021). Opportunity Zones subsidize construction and investment in targeted areas. While there is no central accounting of the number or size of subsidies given by state or local governments for redevelopment projects, with our understanding of the scope of the latter being especially poor, *ad hoc* tabulations indicate instances of state and local redevelopment subsidies number in the hundreds of thousands (Data provided by Good Jobs First, 2022).

Such policies are typically analyzed as place-based policies (Glaeser and Gottlieb, 2008) and rationalized under an equity-efficiency tradeoff. However, the presence of landlord pricing power and resulting redevelopment failure creates a potential efficiency argument for such subsidies. Of note, these subsidies, including those mentioned above, often target or are linked to below-market rents.[19] Our analysis below points to an interesting possibility that, while equity may be the impetus for these, operationally, price restrictions may improve subsidy efficiency in low-information environments.

We first discuss an optimal policy, and then move to describing an implementable welfare improving policy.

Optimal Local Redevelopment Subsidy Eliminating local redevelopment failure can be achieved by aligning the developer's profit function with the local social surplus through subsidizing redevelopment to any quantity $q_{a,1}$ by the change in local renter surplus:

$$\mathsf{S}_a^{\mathsf{Opt}} = [r_a(q_{a,0}) - r_a(q_{a,1})] \cdot q_{a,0} + \int_{q_{a,0}}^{q_{a,1}} r_a(q)\mathrm{d}q. \tag{10}$$

This subsidy is isomorphic to the landlord charging each renter their willingness-to-pay (i.e., perfect price discrimination), which achieves allocative efficiency. Because the planner only cares about total social surplus, the planner accepts 'spending' the entire renter surplus on the landlord if it achieves allocative efficiency. It is unrealistic to believe agents have the information required to implement this subsidy, especially the distribution of willingness-to-pay for individual parcels in the city.[20]

[19]See Inclusionary zoning (Baum-Snow and Marion, 2009; Soltas, 2022) and (Thaden and Wang, 2017).

[20]This optimal subsidy definition assumes all other building rents and quantities remain fixed, and so does *not* account for competition induced benefits, which would require the planner to know additional unobservable information.

Implementable Policies Instead, we consider subsidies targeted at local redevelopment failure that use only information attainable by policy makers. Proposition 1 defines a specific subsidy and rent discount that—when combined—reduce (but do not eliminate) local redevelopment failures and are local social welfare improving, without using any unobservable information.

Policy 1. Provide a redevelopment subsidy $\mathsf{S}_a = C^{\mathsf{d}}(q_{a,0})$ to developer-landlords, i.e. set the subsidy at the reconstruction cost to the building *as-is*.

The intuition for this subsidy is that local redevelopment failure occurs because of the combination of rents on existing structures and positive redevelopment costs. Without the latter (e.g., when no structure exists *ex-ante*), the monopolist always chooses $q^*_{a,1}$. Such a subsidy will always lead to profitable redevelopment. Crucially, these rebuilding costs are already calculated and readily available on a building basis via insurance.

We show in Appendix A that implementing Policy 1 would eliminate local redevelopment failures. However, offering a no-strings-attached subsidy would lead to over utilization such that some redevelopment is local social welfare *dis*-improving (i.e., negative local social welfare change), which we call socially wasteful redevelopment. The city planner cannot *ex ante* tell these cases apart from welfare improving redevelopments. Thus, we consider a second policy to work in tandem with the first.

Policy 2. Require that any developer-landlord accepting a subsidy must set an average rent, $r_{a,\mathsf{S}}$, to satisfy the following rule: $\mathsf{S}_a \leq [r_{a,0} - r_{a,\mathsf{S}}] \cdot q_{a,0}$.

Policy 2 puts an upper bound on the new rent: $r_{a,\mathsf{S}} \leq r_{a,0} - \mathsf{S}_a/q_{a,0}$, and ensures that the local social surplus gain is substantial enough so that subsidies are only taken when socially desirable by future residents. Intuitively, the rule comes from the first term in equation 10, so that the maximal subsidy that is offered is bounded by the local renter surplus. However, because of this bounding, we cannot ensure all local redevelopment failure is eliminated.

Proposition 1. *Policies 1 and 2 jointly are implementable, reduce local redevelopment failure, and are locally social welfare improving.*

The proposition has three consequents. The first is that the policies only require three—observable—market quantities: $\{q_{a,0}, r_{a,0}, C^{\mathsf{d}}(q_{a,0})\}$. The second is that the policies reduce (but do not eliminate) local redevelopment failure. The third is that the policy induced redevelopment generate local social surplus that is greater than the subsidy *and* greater under the policies than the *laissez-faire* outcome. Appendix A proves the claim.

We highlight three points. First, similar to existing policies, Proposition 1 hints at an efficiency argument for pairing redevelopment subsidies with rent reductions. Second, as opposed to existing policies, Proposition 1 prescribes a precise size for both the subsidy and rent reductions with existing information: *ex ante* rents, units, and rebuilding costs (observable on insurance records). Third, because the subsidy is based on rebuilding costs of existing structures, in our framework there is no efficiency argument for development subsidies on empty lots.[21]

3.2 New Implications for Zoning

An immediate implication of the above model is that, even in the absence of spillovers, a policy of no zoning is not first-best. Because a monopolist landlord restricts quantity, the quantity difference between zoning-restricted and an identical, unrestricted parcel with a monopolist landlord is less than the difference between zoning-restricted parcels and a competitively priced parcel. Height *minimums* could reduce rents. Appendix A.8 further explores this point in the context of a monocentric city.

On the other hand to the extent that zoning limits bind at a particular parcel, the zoned quantity is lower than the monopoly-optimal quantity, and rents must be higher than what they would be otherwise. However, in a city where only some parcels are limited by zoning rules, those policy restrictions also impact rents at unzoned parcels by affecting equilibrium pricing power at unconstrained parcels. Intuitively, zoning induced scarcity at one building increases the pricing power at a unzoned rival building. This leads to the following statement:

Proposition 2. *Introducing binding zoning limits on a given parcel increases the markups and rent at all other parcels, including unzoned parcels and parcels where zoning limits do not bind.*

Appendix A presents a proof. Because of pricing power, binding zoning rules that increase rents have spillover effects that increase rents at unconstrained locations. Likewise, relaxing zoning at one parcel brings down rents elsewhere.[22]

[21]We reiterate that both the optimal and implementable policies discussed here pertain to aligning incentives regarding the decision to redevelop, but take the quantity provided by the monopolist conditional on redevelopment, $q^*_{a,1}$ as given. We do not consider policies attempting to set $q_{a,1} > q^*_{a,1}$ such that $P = MC$. Recall that redevelopment failure does not apply to buildings where $q_{a,0} > q^*_{a,0}$, and this policy does not impact those buildings. Importantly, Policy 2 disincentivizes these landlords from requesting the subsidy.

[22]Of course, even when units are priced competitively, if marginal costs are increasing, by limiting supply

Finally, while our model assumes one owner per plot, Appendix A also shows that when this assumption is relaxed, increased concentration via merging ownership of two buildings should increase markups everywhere. Intuitively, an owner of multiple buildings internalizes the cross-price elasticities between those buildings. When a landlord adds a property to a portfolio, she raises rents at all her buildings. And, again because of cross-price elasticities between other buildings, this induces landlords at other buildings to respond by raising rents as well. In Section 5, we will test these predictions.

4 Data

Sources Our main data are derived from public administrative building-level records and scraped data from the New York City departments of City Planning, Finance, and Housing Preservation & Development. We combine the Primary Land Use Tax Lot Output (PLUTO) and the Final Assessment Roll (FAR) for all buildings in NYC. The PLUTO and FAR provide location, zoning, market value, and other building characteristics.

We merge these with data derived from communications between the DOF and building owners, scraped off the Property Tax Public Access web portal, which we call the Notice of Property Value (NPV) dataset. It includes information mailed to building owners including gross revenue and cost estimates and the number of rent stabilized units.

We use the American Community Survey to allocate rental households to buildings to estimate building vacancies.[23] The size of each rental market is the total number of renter households in 4+ unit buildings in the borough.

Sample Our data spans from 2007 to 2019, where we are able to link all datasets together. We use all private buildings classified as multi-family rental buildings in the Bronx, Brooklyn, Manhattan, and Queens with four or more units, where all units are residential units and there is no missing data.[24] Based on Section 2.2, we create an unconstrained subsample that excludes buildings older than 10 years, rent stabilized buildings, and buildings at the zoning limit.[25] Table 1 presents summary statistics for NYC rental buildings for our

at one location, zoning can impact rents and quantities at other locations. But Proposition 2 points out that pricing power exacerbates the price effects by changing optimal markups. In other words, even in a world of constant marginal costs, zoning at one parcel would raise rents at all other parcels in the city by increasing monopoly markups.

[23]We multiply building residential units by the block-group level rental occupancy rate. This assumes vacancy rates are uniform within Census block-groups.

[24]We use these and 'New York City' interchangeably; we omit Staten Island due to its low number of large renal buildings.

[25]We deem a building is rent stabilized if more than 50% of units are rent stabilized, and at the zoning limit if it would not be allowed an additional unit based on FARs and density requirements.

two samples.

Building Rental Income We use scraped data from communications between the city and landlords about building income. In NYC, rental buildings are assessed based on their income generation. The DOF collects annual revenue and cost information for all rental buildings as the basis of a building's tax assessment. Every year, the DOF sends a letter to building owners for each building that includes this information, which we parse for our dataset.[26] We divide building income by the number of units and again by twelve for average monthly rent. A limitation of our data is that we do not see unit-level income.

Other Variables We link buildings using their unique, parcel-specific "borough-block-lot" (BBL) identification codes, with additional verification based on lot characteristics. The building-level characteristics that we include are ten-year building age group indicators, log miles to nearest subway station, log years since the last major building renovation, log average unit square-feet, and whether the building has an elevator, and lot size quartiles. For location controls we use Census tract by year fixed effects.

Table 1: Summary Statistics:
2007-2019 NYC Rental Buildings

	Tract Level	
	Building Level	
	Residential (1)	Unconstrained & New (2)
Median Rent	$ 1059	$ 1199
Median Rent by Median Income	0.18	0.23
Residential Units	19.2	14.6
Years Since Construction	88.2	5.0
Years Since Renovation	69.4	5.0
Avg Unit Sqft	821.5	924.5
Pct w/ Elevator	12%	19%
Miles to Subway	0.3	0.3
Currently Zoning Constrained	64%	0%
Initially Zoning Constrained	77%	12%
Pct Units Rent Stabilized	38%	0%
Pct Built in Last 10 Years	2%	100%
Unique Buildings	58,374	214

Note: The table reports summary statistics for all single-use residential buildings with four or more units and sub-samples. The first column contains all residential, single-use buildings. The second column subsets the data to only include buildings that are less than ten years old, where less than 50% of units are rent-stabilized, and that are able to add an additional unit according to zoning regulations, building floor-area-ratios, and minimum unit area requirements. Building data from PLUTO, NPV, and FAR Monthly rent is building income divided by residential units divided by 12. Median borough income from ACS. All dollar values in nominal terms. Years since construction / renovation equal the year minus the construction year / most recent major renovation year. Avg Unit Sqft is total building area divided by units. Geodesic distances are in log miles based on building coordinates.

[26] Letters inform owners how income and cost information is used to calculate tax liability.

5 Concentration and Rents in New York City

We examine the correlation in the data between ownership concentration and rents. We note that results in this section are not causally identified. However, in line with our extended framework discussed in Appendix A, we find increased concentration is correlated with modest rent increases. .

To examine whether the data are consistent with the predictions of our framework, we first construct ownership shares at the Census tract level, and compare the ten-year change in HHI from 2009 to 2019 on rents to estimate the long-run relationship.[27] Section 4 summarizes the trade offs of tract-level analysis, as well as our construction of tract-level ownership data.

Using our recovered ownership structure, we calculate owners' market share:

$$\mathsf{s}_{g,t}^{f} := \frac{\left(\sum_{j\in\mathcal{A}_{f,g,t}} D_{j,t}\right)}{\sum_{f'\in\mathsf{F}_{g,t}}\left(\sum_{j\in\mathcal{A}_{f',g,t}} D_{j,t}\right)}. \tag{11}$$

where $\mathcal{A}_{f,g,t}$ is the set of j buildings owned by landlord f in tract g in year t, and $\mathsf{F}_{g,t}$ is the set of landlords in that tract and time. Figure 1, plots tract-level HHI measures for NYC, where HHI is the sum of squared owners' shares, $\mathsf{HHI}_{g,t} := \sum_{f'\in\mathsf{F}_{g,t}}\left(\mathsf{s}_{g,t}^{f'}\right)^2$.

We construct a modified "leave-out" HHI index (see Appendix A for theoretical motivation). For each landlord f, we calculate surrounding concentration as the sum of rival hs' market shares:

$$\mathsf{HHI}_{f(j),g,t} := \sum_{h\in\mathsf{F}_{g,t}^{\neg f}}\left(\tilde{\mathsf{s}}_{f,g,t}^{h}\right)^2 := \sum_{h\in\mathsf{F}_{g,t}^{\neg f}}\left(\frac{\left(\sum_{j\in\mathcal{A}_{h,g,t}} D_j\right)}{\sum_{f'\in\mathsf{F}_{g,t}^{\neg f}}\left(\sum_{j\in\mathcal{A}_{f',g,t}} D_j\right)}\right)^2, \tag{12}$$

where $\mathsf{F}_{g,t}^{\neg f}$ is the set of rivals to landlord f. [28]

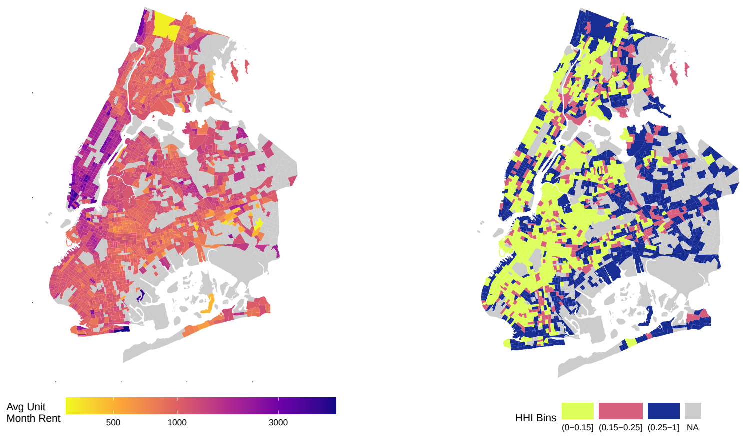
We then test whether rent increases in concentration. Our main specification estimates

$$\ln[\mathsf{r}_{j,g,t}] = \gamma_0 + \gamma_1 \cdot \ln[\mathsf{HHI}_{f(j),g,t}] + \gamma_2 \cdot X_{j,g,t} + \upsilon_{j,g,t}, \tag{13}$$

[27]Given available data, reliable ownership data cannot be constructed before 2009. We calculate concentration off of the full sample of buildings in each year but restrict our sample to single use buildings with matched ownership information.

[28]In Appendix B, we use the more standard construction of HHI and shares in Equation (11). An alternative to using tracts is to estimate location or building-specific markets where competitor's distance-weighted shares are aggregated using distance elasticities derived from LODES commuting flows.

Figure 1: Distribution of New York City Rents & Concentration



Note: The figure displays the geographic distribution of monthly rent and ownership concentration (measured by HHI) for 2010 Census tracts minus Staten Island. The top plot is the 2007-2015 average building monthly rent tract average in 2019 dollars. The bottom plot is the 2008-2015 average tract HHI value; note, according the the US Federal Trade Commission, HHI values between 0.15-0.25 are 'moderately concentrated' and above 0.25 are 'highly concentrated.' Dark blue tracts indicate higher rents (log scale) and concentration. Missing values (in light-gray) are Census tracts where we have insufficient data, in part due to the exclusion of mixed-use buildings. Data from PLUTO, FAR, NPV, and MDRC as described in text.

where $\mathsf{r}_{j,g,t}$ is the average unit rent of building j in tract g at time t, $X_{j,g,t}$ are control variables described below, and υ is an econometric error term.[29] We also include $\ln[\mathsf{s}_{g,t}^{f(j)}]$ in some specifications to test for the impact of owners' shares on rents at their own buildings.

Results Table 2 presents our estimates of equation (12). Columns (1-3) use the log of the leave-out HHI measure and columns (4-6) add the log of the building owner's market share as a control. All columns use year fixed effects.

Columns (1) and (4) include borough fixed effects. We report statistically insignificant negative correlations. However, because these columns treat the data as a repeated cross section and suffer from omitted variable bias, we refrain from interpreting the small and insignificant resulting coefficient. Columns (2) and (5) introduce tract-level fixed effects, which remove unobserved time-invariant tract-level variation like neighborhood quality. In column (2), we estimate that a 10% increase in tract concentration index is associated with a 0.3-0.4% increase in rents across all buildings. Columns (3) and (6), our most stringent specifications, include building-level fixed effects, which remove unobserved time-invariant building characteristics. Similar to columns (2) and (5), a 10% increase in competitors' concentration is associated with a 0.5% increase in rents across all buildings. The coefficient on own share of the tract is inconclusive.

Because market structure is endogenous and we are unable to observe changing tract

[29] While we use general subscripts $\{j, g, t\}$ for $X_{j,g,t}$, in specific specifications some controls will be time invariant; e.g., when using building fixed effects.

Table 2: The Relationship Between Ownership Concentration and Rent

	$\ln$[Average $r_{j,g,t}$]					
	(1)	(2)	(3)	(4)	(5)	(6)
Log Leave-Out Tract HHI	-0.02	0.03	0.05	-0.00	0.03	0.05
	(0.01)	(0.01)	(0.02)	(0.01)	(0.01)	(0.02)
Log Owner Share in Tract				-0.03	-0.01	0.01
				(0.00)	(0.00)	(0.01)
Time-varying controls	Y	Y	Y	Y	Y	Y
Borough FEs	Y	N	N	Y	N	N
Tract FEs	N	Y	N	N	Y	N
Building FEs	N	N	Y	N	N	Y
Observations	63,156	63,087	41,842	63,156	63,087	41,842

Note: The table reports the results from regressions of log of building average unit monthly rent on the log of the 'leave-out' HHI index, calculated at the building level by leaving out the building owner's units. The regression features two time periods, 2009 and 2019, to estimate the long run relationship between tract ownership concentration and rents. Time-varying controls include log building age, log years since renovation, log average unit sqft, and an indicator for having an elevator. Regressions are at the building-year level using all single use rental buildings in New York City. Standard errors are clustered at the Census tract level for all columns.

and building conditions that are correlated with both rents and ownership concentration, we caution against interpreting these coefficients causally, but instead we take reassurance from the concordance between our models theoretical results on concentration (see Appendix A for discussion) and the stylized fact that increases in concentration around a building are correlated with increases in rents.

6 Quantification Exercise: Elasticities and Markups

Our final empirical exercise gauges the competitiveness of our setting by estimating the size of markups (Benassy, 1991). Without imposing assumptions on firm conduct, we estimate demand parameters from our model to generate building OPEs. Using results in Section 2, we isolate the set of buildings for which OPEs provide information on markups.[30]

To identify the demand parameters, we use variation from three sources. As is common in the literature, we use rival characteristics to construct 'BLP instruments' (Bayer, Ferreira, and McMillan, 2007; Davis et al., 2021; Almagro and Domınguez-Iino, 2019; Almagro, Chyn, and Stuart, 2023). To allay endogeneity concerns, we re-estimate using two different cost-shifting instruments with separate sources of variation and identifying assumptions: the tax instrument and historic construction costs. Per our discussion in Section 2 and

[30]Alternatively, we estimate a 'theory-free' OPE using a standard log-log 2SLS regression that is quantitatively similar to the mean values from our structural model (between $[-6.3, -2.7]$). However, we caution that this approach includes zoning-limited and older vintage buildings in a local average treatment effect for which the OPE is not informative for markups.

following Froeb, Tschantz, and Crooke (2003), we note potential demand always equals realized demand, allowing us to proceed in spite of dis-economies of scale.

Empirical Specification Let (j, b, t) index building j in borough b in year t. Building market share and rent are denoted s and r, respectively. We take borough-years as markets. Individuals $i \in \mathsf{N}_{bt}$ choose a unit at a building or an outside option. Parameterizing our model further, renter indirect utility is a linear in rent and amenities:

$$U_{ijbt} = \beta_0 + \beta_1 \cdot X_{jbt} + \alpha_{bt} \cdot \mathsf{r}_{jbt} + \delta_{jbt} + \epsilon_{ijbt}, \tag{14}$$

where δ is a building characteristic that is observed by the market but not in our dataset, $\alpha_{bt} = \alpha/\bar{y}_{bt}$ captures how rent affects utility and is parameterized as a single parameter divided by median borough income, X_{jbt} is a set of observable building characteristics including log distance to the nearest subway station, log average unit square feet, log years since a major renovation, and an indicator for whether a building has an elevator, and a tract-year fixed effects, and ϵ is the individual level idiosyncratic taste shifter (distributed Type-1 Extreme Value). Aggregating individual demand and using the analytic inversion of Berry (1994), we arrive at our estimation equation:

$$\ln[\mathsf{s}_{jbt}] - \ln[\mathsf{s}_{0bt}] = \beta_0 + \beta_1 \cdot X_{jbt} + \alpha_{bt} \cdot \mathsf{r}_{jbt} + \delta_{jbt}, \tag{15}$$

where s_{0bt} is the market share of the borough's 'outside good' (i.e., not choosing a rental property). As in the pass-through model, there is a potential correlation between the rent and the unobserved building characteristic, which leads us to instrument for rent.

Instruments We use three approaches to identify the demand parameters. First, we use the characteristics of rival buildings, commonly called 'BLP instruments.' The degree to which a landlord can markup rent is attenuated when renters have more similar choices. We follow Gandhi and Houde (2018) calculating the sum of squared differences for continuous variables and a count of rivals for indicator variables. We reduce the resulting instruments to a single one following Bayer, McMillan, and Rueben (2004) and Davis et al. (2021).[31] We use the count of all rivals, rivals with elevators, and mixed-use rivals and the sum of square differences in age and distance to closest subway entrance.

The exclusion restriction holds if, controlling for characteristics, building unobservables are orthogonal to rivals' characteristics: $\mathsf{E}[\delta_j \mid X_k] = 0 \ \forall \ j, k$. Conditional on a

[31] We regress rent on the BLP instruments alone and use the fitted value as an instrument for rent in the full estimation equation with controls.

building's own characteristics, rivals' 'closeness' should only affect rent through competition. A concern is that neighborhood characteristics are correlated with rivals' amenities. Accordingly, we omit rivals from 1 km around each building from the instrument construction. Still, the possibility of long-lagged spatial auto-correlation poses an unknowable threat (as throughout this literature), and is the impetus for our second set of strategies.

Second, we construct rents using variation tied to landlord costs using plausibly exogenous idiosyncratic cost variation from assessment procedures, following Watson and Ziv (2025). Third, we use variation in costs at the time of buildings' construction. We match buildings by year of construction to cost indices for unskilled labor and materials cost and NYC real estate tax rates (Barr, 2016), using the three, net of linear cohort trends, as instruments for price.While we place little weight on these results because such costs may be correlated with unobservables, we include this complementary specification because the direction of bias is unlikely to be the same as that of the other instruments.

Markup Calculations with Supply-Side Restrictions While our elasticity estimation is cost agnostic, to derive markups from demand elasticities, we must account for how landlords set rents and quantities in our setting. As discussed in Section 2.4, markups can only be calculated from the OPEs of a subset of buildings: the newly built, policy-unconstrained sample. We only interpret pricing power using results for that subsample.[32]

The logit demand elasticity (Equation 5) is $\varepsilon_{jbt} = \alpha_{bt} \cdot \mathsf{r}_{jbt} \cdot (1 - \mathsf{s}_{jbt})$. If landlords use a Bertrand price competition game to set rents, then the Lerner rule, $\mathsf{L}_{jbt} = (-1/\varepsilon_{jbt})$, reveals the proportion of rent due to markups: $\mathsf{L} = (\mathsf{r} - c)/\mathsf{r}$ for the new, unconstrained sample.

Aggregate Demand Consensus housing demand estimates are inelastic (Chen, Clapp, and Tirtiroglu, 2011; Albouy, Ehrlich, and Liu, 2016). These estimates measure the change in total housing demand with a change in price, rather than the OPEs we target, ε_j. Connecting these previous estimates to ours, we calculate the responsiveness of renters to a 1% increase in rent for all buildings, which we refer to as the "aggregate price elasticity" (Berry and Jia, 2010; Conlon and Gortmaker, 2019):

$$\varepsilon_{bt}^{\text{Agg}} = \sum_{j \in \mathcal{J}_{bt}} \frac{D_{jbt}(\{\mathsf{r}_k + \Delta \mathsf{r}_k\}_{k \in \mathcal{J}_{bt}}) - D_{jbt}}{\Delta}\Bigg|_{\Delta = 1\%}. \tag{16}$$

[32]We reiterate that policy constraints do not bias our estimates. We use the full sample to estimate the demand parameters, which requires only that prices clear demand.

Results Table 3 presents our main empirical results. Column (1) reports a positive, small OLS estimate of α, likely biased due to unobserved amenities. Column (2) reports the 2SLS estimates using BLP instruments. Here α is roughly -13, which corresponds to a median markup on the unconstrained sample of roughly one-third. Our other two specifications, using our synthetic tax instrument (Column 3) and historic cost shifters (Column 4), vary in magnitude, estimating α as roughly -15 to -36, and with a median markup on the relevant sample of 28% and 11%, respectively. One reason for this could be differences in sample.[33] We report first stage effective F statistics (Olea and Pflueger, 2013) and the Anderson-Rubin F statistic for rent. Our three IV specifications broadly agree that α is negative and reject extreme values implying no pricing power. Overall, were units priced at marginal costs reflective of building construction and maintenance, we would expect rents to be about 66-90% of their current levels.

Since monopoly pricing is inconsistent with inelastic demand, observing whether buildings in our sample are elastic is a useful check. Across all 2SLS specifications, all buildings have elastic demand. BLP and Historic Costs specifications estimate *aggregate* elasticities to be roughly -0.6 and -0.7, respectively, within the consensus range of $(-0.7, -0.3)$, while results from the Tax IV are more elastic, -1.7.[34] Figure 2 plots the distribution of the own-price elasticities and markups for each specification.

7 Conclusion

We reconsider the canonical competitive urban model, proposing the existence and quantifying the relevance of markups. We find markups are economically significant and policy relevant. Using plausibly exogenous variation stemming from NYC tax policy changes, we find evidence that multi-family landlords face downward sloping demand from renters. We estimate that markups account for between a tenth and a third of rents. We discuss a welfare improving policy that is implementable without requiring any structural estimates and mirrors widespread inclusionary zoning policies, as well as new spillover from zoning regulations that operates through changes in markups.

These results revise canonical analyses of housing policies and housing markets. They point to markups as a new source of housing supply constraints and markups as a source of spillovers–from supply to demand and between suppliers. More broadly, our analysis

[33] The Tax IV requires observing buildings' base-income in 2007. Imposing the same sample restriction on the BLP generates similar coefficients with the Tax IV, while doing so with the Historic IVs does not change the coefficient. Both estimates become significantly noisier.

[34] Gyourko and Voith (2000); Chen, Clapp, and Tirtiroglu (2011) also find elastic estimates.

Table 3: Demand Estimation Results

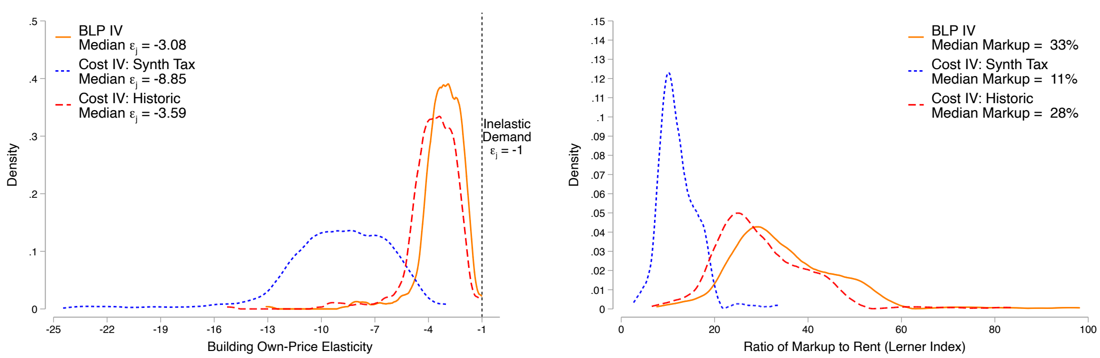
	OLS (1)	BLP (2)	Tax (3)	Historic Costs (4)
α	1.01	-12.72	-36.61	-14.85
	(0.11)	(2.54)	(2.68)	(2.19)
Robust F Stat	-	57.54	233.15	33.50
Robust AR Stat for Rent	-	56.81	1468.04	101.59
Observations	354,435	354,435	183,210	336,139
$\text{Med}(\varepsilon_{jbt})$	0.18	-2.22	-6.40	-2.60
$\text{Med}(\varepsilon_{jbt} \mid \text{Unconst., New})$	0.24	-3.08	-8.85	-3.59
Pct Elastic	0.00%	100%	100%	100%
$\text{Med}(\text{L}_{jbt} \mid \text{Unconst., New})$	0.00	0.33	0.11	0.28
$\text{Avg}(\varepsilon_{bt}^{\text{Agg}})$	0.05	-0.59	-1.70	-0.69

Note: The table displays parameter estimates from Logit demand models. The own-price elasticity is ε, the Lerner rule is $-1/\varepsilon$, and the aggregate price elasticity, ε^{Agg}, is based on Berry and Jia (2010). Buildings are 'unconstrained' if *not* rent stabilized and *not* zoning-constrained; buildings are considered new if less than 10 years old. We estimate an (1) OLS model and three just-identified IV models using (2) 'BLP instruments', (3) the synthetic tax instrument, and (4) historic cost instruments based on the year of building construction. All models include Census tract-year fixed effects, controls for log distance to nearest subway station, building age, years since renovation, average unit square-feet, and an indicator for having an elevator. Standard errors are clustered by Census tract and the first stage F statistic and the Anderson-Rubin F statistic for the estimated coefficients are cluster robust as well.

Figure 2: Distribution of Results

(a) Own-Price Elasticity

(b) Markup as a Percent of Rent



Note: The figure plots the kernel density plot of own-price elasticities (Panel (a)) and markups / Lerner rule (Panel (b)) based on reults from Table 3. The solid orange line uses the BLP IVs (column 2), the short-dash blue line uses the Cost IV using counterfactual tax per unit (column 3), and the long-dash red line uses the Cost IV using historic building costs (column 4). The sample is all single-use residential buildings in the four boroughs with with four or more units, where unconstrained means all buildings that are not at the zoning limit and where less than 50% of units are rent stabilized and where new buildings are 10 years old or less. The vertical line in Panel (a) indicates elasticities greater than -1, which would be incompatible with monopolistic pricing; these elasticities are excluded from Panel (b). The models and estimation are described in the text.

implies markups should be directly modeled in empirical work geared towards estimating housing supply curves, counterfactual re-allocations, and estimates of the impact of restrictive housing policies.

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Online Appendix of
Is the Rent Too High? Land Ownership and Monopoly Power

C. Luke Watson and Oren Ziv

A Theoretical Appendix

In this section, we provide proofs of and expanded analyses of our theoretical results.

A.1 Alternatives to Maggi (1996) Cost Function

A.1.1 Current Approach

In Maggi (1996), duopolists set capacity and prices and can service demand above capacity at a potentially arbitrarily high marginal cost. Benassy (1991) points out that these types of dis-economies of scale do not on their own prevent rationing. Our oligopolistic setting differs from Maggi (1996), as well because of additional dis-economies of scale in capacity provision. However the same logic outlined in Wauthy et al. (2014)–that Maggi's equilibrium rests on an effective ban on rationing that forces competitors to service (hypothetical) demand even if it were unprofitable to do so–extends to our setting in a straightforward way. The threat of competitors' unprofitable quantity provision at posted prices returns each player's residual demand to that which would be the case if capacity constraints did not bind at any off-equilibrium price, and therefore the pricing sub-game should have a pure strategy equilibrium. Indeed Greenfield and Sandford (2018) use the same setup in a similar oligopoly with asymmetric costs.

The palatability of these assumptions for our setting is mixed. Lessors do face short-run constraints. However the division between the long and short run is not as rigid as is expressed in our model via timing assumptions on the capacity and pricing stages. Zoning constraints also could be viewed either as a rigid constraint that bind for developers, or a dis-economy of scale imposed by policy, which can and often are relaxed through payments in time, effort, and side payments.

A.1.2 Alternative Approaches

The pursuit of a pure-strategy equilibrium solution for differentiated goods markets with capacity constraints has been elusive. Yin and Ng (1997), as amended by Schulz (2000) and Yin and Ng (2000), provide a limited context with linear demand where a pure strategy equilibrium does exist. These results seem encumbered by functional form assumptions, which we feel hamper their effective application in our context.

Schulz (1999) offers an alternative approach: self-confirming beliefs of consumers that they will not be rationed lead them to choose buildings facing capacity constraints. When deviations in competitor prices do induce rationing, they are prevented from returning to that competitor's product by search costs. Crucially, if a deviation in competitor pricing occurs, consumers are prohibited from forming new beliefs about rationing. Presumably, search costs do not occur in the initial search. Under linear demand, arbitrarily high, finite search costs are sufficient to preclude re-searching. This is not the case with our logit demand under all rationing rules, but is with efficient rationing. Two conditions are needed to avoid a kink in demand: first, an increase in price must induce the same change in initial consumer building selection as would be the case without rationing, and second, rationed consumers must not be allowed to return to the deviating building. The first is achieved with self-confirming beliefs. The second is achieved with search costs.

A.1.3 After-market Arbitrage

Another possible solution is allowing for an after-market arbitrage between consumers who were allocated spots in rationed buildings and those who attempted to gain entry but were not allotted spots. In this sub-game setup, prices are posted, consumers apply for buildings in any order they choose, facing an increasing marginal cost of application (which could be pecuniary or time costs). After applying to a building, spots are allocated by lessors at random. If they are not allocated a spot, they can search again or participate in an after-market with the mass of consumers who were allocated spots, exchanging their spot for some endogenously determined market-clearing price. There is no search cost for the outside option.

In contrast to Schulz, here consumers know demand, prices and capacity, so they are aware that they face a lottery if they choose certain buildings. Off-equilibrium, their demand for such a building includes the payoff if they win, whether they choose to stay or sell, the payoff if they lose, and the probability of each. The secondary market is competitive because consumers are small, so those who choose to sell will have the lowest willingness to pay for the rationed building. This ensures a finite market clearing price. If the second search costs are chosen to be higher than that price, no one will choose to enter the lottery for a constrained building only to sell to return to an unconstrained option, except for those who initially prefer the outside option.

The key is that in this setup, at the consumer's initial choice, the outside option is always dominated by the choice of entering the lottery for a constrained building. Winning the lottery for a constrained building effectively lowers the price of the outside option for those consumers who prefer the outside option in a "traditional equilibrium" without capacity constraints. This increases the share of consumers choosing the constrained building (in lieu of the outside option) in the first search. Raising prices at unconstrained buildings does two things: it reduces the elasticity of residual demand by inducing rationing at constrained buildings, making those choices relatively less attractive, but reduces the cost of the outside option, increasing elasticity. When the share of the outside option is large (or, in terms of model primitives, its mean value arbitrarily high), the latter effect dominates and the demand curve kinks in the opposite direction, so raising prices results in more elastic demand.

A.2 Local Redevelopment Failure Details

We claimed that given any set of costs, we could construct a demand schedule such that redevelopment failure occurs. Here, we show this in four steps. Note, we ignore 'landlord services' costs, $C^{\dagger}$, for ease of notation.

First, choose any development cost function $C : \mathcal{R} \to \mathcal{R}$ such that $C(\cdot)$ is continuous, convex, and strictly increasing. Now, fix some initial quantity: q^0.

Second, we choose an inverse demand function $r : \mathcal{R} \to \mathcal{R}$ such that $r(\cdot)$ is continuous, differentiable, and strictly decreasing. We also choose this demand function, given $C(\cdot)$ and q^0, such that it satisfies three properties: [1] $q^* = \arg\max\{r(q) \cdot q - C(q)\}$, [2] $q^0 < q^*$, and [3] $r(q^*) \cdot q^* - C(q^*) > r(q^0) \cdot q^0$. From these properties, $r(q^*) \cdot q^* - C(q^*) > r(q^*) \cdot q^0$ as $r(q^*) < r(q^0)$. We can rearrange the inequality: $r(q^*)(q^* - q^0) - \int_{q^0}^{q^*} C'(q)\mathrm{d}q - C(q^0) > 0$. By assumptions on $r(\cdot)$: $\int_{q^0}^{q^*} r(q)\mathrm{d}q > r(q^*)(q^* - q^0)$,[35] and so $\int_{q^0}^{q^*} (r(q) - C'(q))\,\mathrm{d}q > C(q^0)$.

So far, from an arbitrary cost function and initial quantity, we created an inverse demand function such that redevelopment is both profitable and socially desirable.

Third, we consider a new inverse demand function, $\tilde{r}(q) = r(q) - A$, where $A \in \mathcal{R}$. For any $A \neq 0$, this new demand schedule creates to a new initial rent, $\tilde{r}(q^0)$, and a new monopoly point, $(q^A, \tilde{r}(q^A))$. Thus, manipulating the A term allows us to adjust the social surplus of redevelopment.

Fourth, we choose a specific A such that $\int_{q^0}^{q^S} (\tilde{r}(q) - C'(q))\,\mathrm{d}q = C(q^0)$. That is, we use A to vertically shift the inverse demand down until the the area under its curve is equal to the fixed amount $C(q^0)$. Thus, redevelopment is now marginally socially desirable. However, by definition of the integral and the demand assumptions: $\tilde{r}(q^S)(q^S - q^0) < C(q^S)$, so redevelopment is no longer profitable to the landlord. □

[35] That is, the area under a continuous curve (with negative slope) is greater than the rectangle within it.

A.3 Policy 1 Eliminates Local Redevelopment Failure

Here we prove our claimed that Policy 1 eliminates local redevelopment failure. Suppose that $\int_{q^0}^{q^*}(r(q)-c(q))\mathrm{d}q - C(q^0) > 0$ but $r(q^*)\cdot q^* - C(q^*) < r(q^0)\cdot q^0$, so that redevelopment is socially optimal but privately not profitable. Redevelopment failure is solved if the subsidy, S, causes redevelopment to be net-profitable: $r(q^*)\cdot q^* - C(q^*) + \mathsf{S} - r(q^0)\cdot q^0 > 0$. Recall that $q^* = \arg\max\{r(q)\cdot q - C(q)\}$, so by definition $r(q^*)\cdot q^* - C(q^*) > r(q^0)\cdot q^0 - C(q^0)$. If $\mathsf{S} = C(q^0)$, then we have satisfied the claim. □

A.4 Proof of Proposition 1

Proposition 1 has three consequents. The first is that the policy is implementable using only observable market information. The second is that local redevelopment failure is reduced. These two consequents are obvious from inspection.

The third is that the policies together are local social welfare improving. This claim implicitly has two parts: first, that the subsidies are never larger than the social surplus created from redevelopment (no socially wasteful expenditure); second, that the policies lead to positive social welfare relative to *laissez-faire* outcome (we never distort privately optimal redevelopments). Below, we prove this claim, again ignoring $C^{\mathsf{f}}(\cdot)$ for notational ease.

A.4.1 Subsidized Redevelopment is Welfare Improving

Recall, Policy 2 introduces the following rule: $q\cdot(r - r^{\mathsf{S}}) \geq \mathsf{S} = C(q^0)$. This rule states that the landlord must commit to a specific lower rent such that $r^{\mathsf{S}} \leq r - C(q^0)/q^0$. To show that this rule ensures no socially wasteful expenditure occurs, we specify three cases. Let q^{S} be the chosen quantity by the landlord under the subsidy regime. Once rent is fixed by Policy 2 and the landlord is guaranteed at least the subsidy amount, we allow for the possibility of an arbitrary choice of quantity as long as it is revealed as profitable to the landlord.

Case One: $r^{\mathsf{S}} \geq r(q^*)$ — In this case, the rent commitment is not binding, so the landlord will choose the profit maximizing allocation $(q^*, r(q^*))$. The social surplus of this allocation is $\int_{q^0}^{q^*}(r(q) - C'(q))\,\mathsf{d}q > 0$. We know that:

$$\int_{q^0}^{q^*}(r(q) - C'(q))\,\mathsf{d}q > (r(q^*) - C'(q^*)) \cdot \left(q^* - q^0\right) > \left(r(q^0) - r(q^*)\right) \cdot \left(q^0\right) = C(q^0). \tag{17}$$

The first inequality is by assumptions on functional forms (demand is convex decreasing, cost is convex increasing), the second inequality is by profit maximization, and the equality is the definition of subsidy. Thus, the net social surplus is greater than the subsidy.

Case Two: $r^{\mathsf{S}} < r(q^*)$ & $q^{\mathsf{S}} \geq q^*$ — We note that the landlord will never choose to set q^{S} such that the rent is less than the marginal cost; else, the landlord faces a net loss on these units. Next, we note that from Case One redevelopment to q^* is socially desired, and that the point where marginal cost equals rent is the allocation where social surplus is maximized: $(q^{\mathsf{Opt}}, r(q^{\mathsf{Opt}}))$. By continuity and strict monotonicity of $r(\cdot)$ and $C(\cdot)$, it must be that net social surplus is positive for all $q^{\mathsf{S}} \in (q^*, q^{\mathsf{Opt}})$. In fact, net social surplus is greater when $q^{\mathsf{S}} > q^*$.

Case Three: $r^{\mathsf{S}} < r(q^*)$ & $q^{\mathsf{S}} < q^*$ — In this case, we rely on the revealed preference of the landlord to take the subsidy versus the *status quo*; i.e., that the subsidized profit with redevelopment is greater than leaving the building as-is.

First, the landlord will never choose to set $q^{\mathsf{S}} < q^0$. To see this, consider the case where $q^{\mathsf{S}} = q^0 - e$, $e > 0$. The subsidized profit condition yields:

$$r^{\mathsf{S}} \cdot (q^0 - e) - C(q^0 - e) + C(q^0) > r^0 \cdot q^0 \tag{18}$$

$$q \cdot (r^{\mathsf{S}} - r^0) - e \cdot r^{\mathsf{S}} - \int_{q^0 - e}^{q^0} c(q)\mathsf{d}q > 0 \;\; \rightarrow\leftarrow . \tag{19}$$

Thus, $q^{\mathsf{S}} \in (q^0, q^*)$.

Next, to show that the social surplus is still be greater than the subsidy, we again use the profit condition:

$$
\begin{aligned}
& r^{\mathsf{S}} \cdot q^{\mathsf{S}} - C(q^{\mathsf{S}}) + C(q^0) > r^0 \cdot q^0 && (20)\\
\implies & r^{\mathsf{S}} \cdot q^{\mathsf{S}} - C(q^{\mathsf{S}}) > r^0 \cdot q^0 - C(q^0) = r^{\mathsf{S}} \cdot q^0 && (21)\\
\implies & r^{\mathsf{S}} \cdot (q^{\mathsf{S}} - q^0) - C(q^{\mathsf{S}}) > 0 && (22)\\
\implies & r^{\mathsf{S}} \cdot (q^{\mathsf{S}} - q^0) - \int_{q^0}^{q^{\mathsf{S}}} c(q)\mathsf{d}q - C(q^0) > 0 && (23)\\
\implies & r^{\mathsf{S}} \cdot (q^{\mathsf{S}} - q^0) - \int_{q^0}^{q^{\mathsf{S}}} c(q)\mathsf{d}q > C(q^0) && (24)\\
\implies & \int_{q^0}^{q^{\mathsf{S}}} r(q)\mathsf{d}q - \int_{q^0}^{q^{\mathsf{S}}} c(q)\mathsf{d}q > C(q^0) = \mathsf{S}. && (25)
\end{aligned}
$$

Thus, subsidized redevelopment to $q^{\mathsf{S}} \in (q^0, q^*)$ is still socially desirable given our rent commitment rule and that the landlord selects into redevelopment.

A.4.2 No Private Redevelopment is Distorted

We have just shown that any redevelopments that are subsidized must be locally welfare improving relative to the initial building. Here, we show that under Policies 1 and 2, if redevelopment is privately profitable without a subsidy, then the subsidy does not distort this decision towards less development.

Consider three allocations: 'initial' (r_0, q_0), 'monopoly' (r_1, q_1), and 'subsidy' $(r_{\mathsf{S}}, q_{\mathsf{S}})$. The monopoly point is a *laissez-faire* profit maximizing allocation, and the subsidy point is the profit maximizing allocation under the policies. For notational ease, we denote terms like profit, marginal cost, and average as π_j, c_j, a_j for $j \in \{0, 1, \mathsf{S}\}$.

Recall that $\pi_1 = (r_1 - a_1) \cdot q_1$, $\pi_0 = r_0 \cdot q_0$, and $\pi_{\mathsf{S}} = (r_{\mathsf{S}} - a_{\mathsf{S}}) \cdot q_{\mathsf{S}} + \mathsf{S}$, where $\mathsf{S} = a_0 \cdot q_0$. As before, we ignore costs from landlord services to focus on the redevelopment decisions. Next, recall that Policy 2 is that $(r_0 - r_{\mathsf{S}}) = a_0$ or equivalently $r_{\mathsf{S}} = r_0 - a_0$. Finally, note that our logit demand function is convex / concave-up in price, which implies that the inverse demand function is convex / concave-up in quantity, so $\partial^2 r / \partial q^2 > 0$.

We want to show that under Policies 1 and 2, $\pi_1 - \pi_0 \geq 0 \implies \neg\{q_{\mathsf{S}} \leq q_1 \wedge \pi_1 - \pi_{\mathsf{S}} \leq 0\}$. The proof of this is shown in two cases.

Case One: suppose that $c_a \leq a_{\mathsf{S}}$. Then:

$$\pi_1 - \pi_{\mathsf{S}} = (r_1 - a_1) \cdot q_1 - [(r_{\mathsf{S}} - a_{\mathsf{S}}) \cdot q_{\mathsf{S}} + \mathsf{S}] \tag{26}$$

$$= (r_1 - a_1) \cdot q_1 - [(c_{\mathsf{S}} - a_{\mathsf{S}}) \cdot q_{\mathsf{S}} + a_0 \cdot q_0] \tag{27}$$

$$> (r_1 - a_1) \cdot q_1 - a_0 \cdot q_0 \tag{28}$$

$$> (r_1 - a_1) \cdot q_1 - r_0 \cdot q_0 = \pi_1 - \pi_0 \geq 0. \tag{29}$$

Case Two: suppose that $c_a > a_{\mathsf{S}}$. We introduce four minor results. First is Lemma 1:

Lemma 1. *If $c(q) - a(q) \geq 0$, then $(c(q') - a(q')) \geq (c(q) - a(q))$ for all $q' \geq q$, and is strict if $q' > q$.*

This can be seen through differentiation and noting that $C^{\mathsf{d}}(q)$ is convex. Lemma 1 implies that $(c_1 - a_1) \geq (c_{\mathsf{S}} - a_{\mathsf{S}})$ and is strict if $q_1 > q_{\mathsf{S}}$. Next, Lemma 2 states that the Policy 2 rent is based on the marginal cost curve:

Lemma 2. *$r_{\mathsf{S}} = c(q_{\mathsf{S}})$.*

Recall that r_{S} is fixed, so the marginal revenue is perfectly elastic and equal to r_{S}.[36] Next, Lemma 3 describes profit condition $(\pi_1 - \pi_{\mathsf{S}})$ as a function of q_{S}:

Lemma 3. *$(\pi_1 - \pi_{\mathsf{S}})$ is continuous and strictly decreasing in q_{S}.*

Define $\Psi(q_{\mathsf{s}}) = (\pi_1 - \pi_{\mathsf{S}}) = A - [c(q_{\mathsf{S}}) - a(q_{\mathsf{S}})] \cdot q_{\mathsf{S}}$, and so:

$$\frac{\partial \Psi}{\partial q_{\mathsf{S}}} = -[(c'(q_{\mathsf{S}}) - a'(q_{\mathsf{S}})) \cdot q_{\mathsf{S}} + (c_{\mathsf{S}} - a_{\mathsf{S}})] < 0. \tag{30}$$

Finally, using Lemma 3, we can state the following:

Lemma 4. *If $q_{\mathsf{S}} = q_1$ and $\Psi(q_{\mathsf{S}}) > 0$, then $\Psi(q_{\mathsf{S}}) > 0$ for all $q_{\mathsf{S}} < q_1$.*

Thus, to prove our claim, we only need to show that, along with other assumptions for this case, $q_{\mathsf{S}} = q_1$ implies $\Psi(q_{\mathsf{S}}) > 0$. Note that $q_{\mathsf{S}} = q_1$ implies the following:

1. $c_{\mathsf{S}} = c_1 > a_1 = a_{\mathsf{S}}$, and

2. $r_{\mathsf{S}} = (r_0 - a_0) \implies a_0 = (r_0 - c_1)$.

[36] Thus, if $r < c$, then by reducing q profit goes up since the additional units lose revenue; if $r > c$, then by increasing q profit goes up since additional units earn revenue.

Now, see that:

$$
\begin{aligned}
(\pi_1 - \pi_{\mathsf{S}}) &= (r_1 - a_1)\cdot q_1 - [(r_{\mathsf{S}} - a_{\mathsf{S}})\cdot q_{\mathsf{S}} + \mathsf{S}] && (31)\\
&= (r_1 - a_1)\cdot q_1 - [(c_1 - a_1)\cdot q_1 + (r_0 - c_1)\cdot q_0] && (32)\\
&= [(r_1 - c_1) + (c_1 - a_1)]\cdot q_1 - [(c_1 - a_1)\cdot q_1 + (r_0 - c_1)\cdot q_0] && (33)\\
&= (r_1 - c_1)\cdot q_1 - (r_0 - c_1)\cdot q_0 && (34)\\
&= (r_1 - c_1)\cdot (q_1 - q_0) - (r_0 - r_1)\cdot q_0. && (35)
\end{aligned}
$$

Thus $(\pi_1 - \pi_{\mathsf{S}}) \geq 0 \iff (r_1 - c_1)\cdot(q_1 - q_0) - (r_0 - r_1)\cdot q_0 \geq 0$. We next manipulate the right hand inequality:

$$
\begin{aligned}
&\iff (r_1 - c_1)\cdot(q_1 - q_0) \geq (r_0 - r_1)\cdot q_0 && (36)\\
&\iff \frac{(r_1 - c_1)}{r_1}\cdot r_1 \geq \frac{(r_0 - r_1)}{r_0}\cdot\frac{q_0}{(q_1 - q_0)}\cdot r_0 && (37)\\
&\iff \frac{-r_1}{\varepsilon_1} \geq \frac{-r_0}{\frac{(q_1-q_0)}{(r_1-r_0)}\frac{r_0}{q_0}} && (38)\\
&\iff \frac{-r_1}{\frac{\partial q}{\partial r}\Big|_{q_1}\frac{r_1}{q_1}} \geq \frac{-r_0}{\frac{(q_1-q_0)}{(r_1-r_0)}\frac{r_0}{q_0}} && (39)\\
&\iff \frac{-q_1}{\frac{\partial q}{\partial r}\Big|_{q_1}} \geq \frac{-q_0}{\frac{(q_1-q_0)}{(r_1-r_0)}} && (40)\\
&\iff \frac{q_1}{q_0} \geq \frac{-\frac{\partial q}{\partial r}\Big|_{q_1}}{-\frac{(q_1-q_0)}{(r_1-r_0)}}. && (41)
\end{aligned}
$$

Finally, we note that $\partial^2 r/\partial q^2 \geq 0$ implies $\frac{q_1}{q_0} \geq 1 \geq \frac{-\frac{\partial q}{\partial r}\big|_{q_1}}{-\frac{(q_1-q_0)}{(r_1-r_0)}}$. That is, since the first derivative is negative, a positive second derivative implies the absolute value of the first derivative is getting smaller (towards zero from negative infinity). It is relatively straight-forward to show for convex / concave up functions that:

$$
\mathsf{abs}\left(\frac{q' - q}{r' - r}\right) \geq \mathsf{abs}\left(\frac{\partial q}{\partial r}\bigg|_{q'}\right) \qquad \text{if} \quad q' > q \quad \& \quad \partial q(r)/\partial r < 0. \qquad (42)
$$

Thus, we have shown that no privately optimal redevelopment is distorted. □

A.5 Proof of Proposition 2

Binding zoning restrictions, by reducing quantities at a given plot, increase rents at that plot. The rest of Proposition 2 will follow as long as plots, as competing products, are strategic complements in pricing decisions.

Definition A.1. Strategic Complements: If the cross derivative of a given player's own payoff function with respect to her action and that a rival's action is positive, then the actions are strategic complements.

Here, we switch to indexing buildings using j and k rather than the building quality, a. Recall that $D_j = \frac{\mathrm{e}^{F(a_j, r(a_j))}}{\sum_{k\in\mathcal{A}}\{\mathrm{e}^{F(a_k, r(a_k))}\}}$ from the main text using logit demand. To make notation easier, define $\alpha = \partial F(a,r)/\partial r < 0$ be the (negative) marginal utility of consumption. In addition, denote $C(q)$ as the composite cost function and likewise $c(q)$ for marginal cost.[37]

In our Bertrand oligopoly setting, rents are strategic complements if

$$\frac{\partial^2 \pi_j}{\partial r_j \partial r_k} = \frac{\partial\left[\partial D_j/\partial r_j\right]}{\partial r_k}\cdot(r_j - c_j) + \frac{\partial D_j}{\partial r_j}\cdot\left(-\frac{\partial c_j}{\partial q}\frac{\partial D_j}{\partial r_k}\right) + \frac{\partial D_j}{\partial r_k} \geq 0. \tag{43}$$

When we apply Logit demand functions, this becomes:

$$\frac{\partial^2 \pi_j}{\partial r_j \partial r_k} = -\alpha^2 D_j D_k(1-2D_j)(r_j - C_j) - c_j \alpha D_j (1-D_j) - \alpha D_j D_k \tag{44}$$

$$= \underbrace{-\alpha D_j D_k}_{>0}\left[\underbrace{\frac{D_j}{(1-D_j)}}_{>0} + \underbrace{(-c_j \alpha D_j (1-D_j))}_{>0 \text{ only if } c_j>0}\right]. \tag{45}$$

A sufficient condition for strategic complements in the logit case is that $c_j \geq 0 \forall j$. This is true with constant marginal costs or diseconomies of scale for the building. With decreasing marginal costs, the strategic complementary of pricing decisions is ambiguous and may vary between pairs of buildings.

If marginal cost is constant, then the rent increase could only be due to an increase in monopoly markups. With variable marginal cost, the rent changes are a mix of markup and marginal cost increases. Decreasing marginal costs would push the landlord to expand quantity supplied and travel further down the demand curve, which may lead

[37] Alternatively, assume all redevelopment decisions are fixed, and landlords are only competing on occupancy and considering their landlord service costs.

to a smaller markup per unit but greater profit (and lower rent). On the other hand, increasing marginal costs attenuate the landlord's desire to expand keeping the landlord in a steeper part of the demand curve but with greater marginal costs eating into the markup.

If marginal cost is 'locally constant' in equilibrium (i.e., its change is 'small enough'), then we can say buildings are strategic complements in the logit case. Given strategic complements of price strategies, an increase in zoning constrained building k's rent will increase demand for unzoned building j, and increases the price at j accordingly. $\square$

A.6 Discussion of impact of Concentration.

In our main text, each plot is owned by a single agent at all times. Here, we discuss the effects of changes in building ownership. We prove that when an landlord's parcel ownership concentration increases, the landlord increases the prices at all properties. Recall, that consider only the joint developer-landlord problem and relax the assumption that each landlord owns only one parcel. We apply the framework of Nocke and Schutz (2018a,b) to calculate the price effect by utilizing the 'ι-markup of the landlord' from the authors' papers. While those authors use a nested-logit model with constant marginal cost, we remove the nesting structure and allow variable marginal cost.

To prove the proposition, we wish to show that, in the logit case with non-decreasing marginal cost, rent is increasing in market-share of the landlord: $\frac{\partial r_j}{\partial s_f} > 0, \ \forall \ j \in f$. Below, we show this in the two product case for intuition and then in the general case with an arbitrary number of buildings. As in the proof above, we use a composite cost function $C(q)$ and its marginal cost $c(q)$.

Oligopolist Pricing Equation First, we show that landlord f chooses a common markup (Nocke and Schutz, 2018a,b). Let each landlord solves the following joint-profit equation:

$$\max_{\{r_j\}_{j\in f}} \sum_{j\in f} r_j D_j(r_j) - C_j(D_j(r_j)). \tag{46}$$

Following the insight from Nocke and Schutz (2018b), the first order for each property satisfies:

$$(r_j - c_j) = \frac{-1}{\alpha} + \pi_f = \frac{-1}{\alpha(1 - s_f)} \implies r_j = c_j - \frac{1}{\alpha(1 - s_f)} > 0, \tag{47}$$

where $\alpha < 0$ the (negative) marginal utility of consumption. We assume that $c_j > 0$ and its derivative is positive: $\tilde{c}_j := \frac{\partial c_j}{\partial q} \geq 0, \ \forall j \in J$.[38]

Two Product Case Recall again that under logit demand:

$$\frac{\partial D_j}{\partial r_j} = \alpha D_j(1 - D_j) < 0 \tag{48}$$

$$\frac{\partial D_k}{\partial r_j} = -\alpha D_j D_k > 0 \tag{49}$$

Now, we differentiate with respect to the landlord's total market share:

$$r_j = \frac{-1}{\alpha(1 - s_f)} + c_j \tag{50}$$

$$\implies \frac{\partial r_j}{\partial s_f} = \frac{-1}{\alpha(1 - s_f)^2} + \frac{\partial c_j}{\partial q}\left(\frac{\partial D_j}{\partial r_j}\frac{\partial r_j}{\partial s_f} + \frac{\partial D_j}{\partial r_j}\frac{\partial r_k}{\partial s_f}\right) \tag{51}$$

and by symmetry:

$$\frac{\partial r_j}{\partial s_f} = \frac{\frac{-1}{\alpha(1-s_f)^2} + \frac{\partial c_j}{\partial q}\frac{\partial D_j}{\partial r_k}\left[\frac{\frac{-1}{\alpha(1-s_f)^2} + \frac{\partial c_k}{\partial q}\frac{\partial D_k}{\partial r_j}\frac{\partial r_j}{\partial s_f}}{\left(1 - \frac{\partial c_k}{\partial q}\frac{\partial D_k}{\partial r_k}\right)}\right]}{\left(1 - \frac{\partial c_j}{\partial q}\frac{\partial D_j}{\partial r_j}\right)} \tag{52}$$

$$= \frac{-1}{\alpha(1 - s_f)^2}\left[\frac{1 - \frac{\partial c_k}{\partial q}\frac{\partial D_k}{\partial r_k} + \frac{\partial c_j}{\partial q}\frac{\partial D_j}{\partial r_k}}{\left(1 - \frac{\partial c_k}{\partial q}\frac{\partial D_k}{\partial r_k}\right)\left(1 - \frac{\partial c_j}{\partial q}\frac{\partial D_j}{\partial r_j}\right) - \left(\frac{\partial c_j}{\partial q}\frac{\partial D_j}{\partial r_k}\right)\left(\frac{\partial c_k}{\partial q}\frac{\partial D_k}{\partial r_j}\right)}\right]. \tag{53}$$

Finally, we impose the logit model to get:

$$\frac{-1}{\alpha(1 - s_f)^2}\left[\frac{1 - \frac{\partial c_k}{\partial q}\frac{\partial D_k}{\partial r_k} + \frac{\partial c_j}{\partial q}\frac{\partial D_j}{\partial r_k}}{1 - \frac{\partial c_k}{\partial q}\frac{\partial D_k}{\partial r_k} - \frac{\partial c_j}{\partial q}\frac{\partial D_j}{\partial r_j} - \frac{\partial c_j}{\partial q}\frac{\partial c_k}{\partial q}\frac{\partial D_k}{\partial r_j}\alpha(1 - s_f)}\right] > 0. \tag{54}$$

[38] A micro-foundation is that the residential space production function is concave in inputs which implies that the cost function in convex in quantity; hence, marginal cost is non-decreasing in quantity.

A.7 General Product Case

Note that we have the following:

$$[r_i] = [\Gamma(s_f) \cdot 1_f] + [c_i(D_i)] \tag{55}$$

$$\mathsf{D}_{s_f} r = [\Gamma'(s_f) \cdot 1_f] + \mathsf{D}_q c \cdot \mathsf{D}_r D \cdot \mathsf{D}_{s_f} r \tag{56}$$

$$\implies \mathsf{D}_{s_f} r \cdot [\mathbb{I} - \mathsf{D}_q c \cdot \mathsf{D}_r D] = [\Gamma'(s_f) \cdot 1_f] \tag{57}$$

$$\implies \mathsf{D}_{s_f} r = [\mathbb{I} - \mathsf{D}_q c \cdot \mathsf{D}_r D]^{-1} \cdot [\Gamma'(s_f) \cdot 1_f] \tag{58}$$

A.7.1 Definitions and Lemmas

Definition A.2. Strictly (Row) Diagonally Dominant : for every row, i, the element along the diagonal, a_{ii}, is greater in magnitude than the sum of the magnitudes of each non-diagonal element in the row $a_{i,j},\ j \neq i$. That is, $|a_{i,i}| > \sum_{j \neq i} |a_{i,j}|$.

Definition A.3. $\mathtt{Z}$-matrix : a matrix whose off-diagonal entries are less than or equal to zero.

Definition A.4. $\mathtt{M}$-matrix : a $\mathtt{Z}$-matrix where every real eigenvalue of A is positive.

Lemma 5. *If A is a $\mathtt{Z}$-matrix that is strictly diagonally dominant, then A is an $\mathtt{M}$-matrix by* Gershgorin Circle Theorem.

Lemma 6. *If A is an $\mathtt{M}$-matrix with positive diagonals and negative off diagonals, then $B = A^{-1}$ is monotone positive; i.e., $b_{ij} > 0,\ \forall\ i, j$; proof in Fan 1958.*

A.7.2 General Case Proof

We need to show that the lemma holds and that the vector $B \cdot \Gamma'(s)$ is a monotone positive vector. Let $[\mathbb{I} - \mathsf{D}_q c \cdot \mathsf{D}_r D] = A$.

First, see that A is (a) a Z-matrix that is (b) Strictly (Row) Diagonally Dominant :

(a) for each row, using logit demand, we have

$$a_{i,i} = 1 - \tilde{c}_i \alpha D_i (1 - D_i) > 0 \tag{59}$$

$$a_{i,j} = \tilde{c}_i \alpha D_i D_j < 0 \tag{60}$$

(b) plug into definition of (row) diagonally dominant

$$\Longrightarrow 1 + \tilde{c}_i |\alpha| D_i (1 - D_i) > \sum_{j \in f \setminus i} \tilde{c}_i |\alpha| D_i D_j = \tilde{c}_i |\alpha| D_i \sum_{j \in f \setminus i} D_j \tag{61}$$

$$\Longrightarrow 1 + \tilde{c}_i |\alpha| D_i > \tilde{c}_i |\alpha| D_i \cdot s_f. \tag{62}$$

Thus A satisfies lemma 6, so B is a monotone positive matrix.
Second, $\Gamma'(s_f) = \frac{\mathsf{d}}{\mathsf{d}s_f} \frac{-1}{\alpha(1-s_f)} = \frac{-1}{\alpha(1-s_f)^2} > 0$.
Thus as $B \cdot \Gamma'(s_f)$ is a series of multiplication and addition of positive numbers, so $\mathsf{D}_{s_f} r$ must be a monotone positive vector. $\square$

A.8 Alternative Models

In this section, we explore how alternative frameworks affect theoretical results. We first explore a competitive model as an instructive benchmark relative to the model in the main text. We then explore several extensions of our model.

A.8.1 Competitive Urban Model

Here, we develop an urban model where rents are differentiated based on local amenities but landlords still have no pricing power. The primary assumptions are that landlords view themselves as atomistic and that renters' preferences are uniform.

Consider a city with fixed population of renters $i \in N$ and a set of locations $j \in J$. Each location has a local amenity x_j and rent r_j, and is owned by a unique, atomistic landlord who is a price-taker providing housing with a marginal cost of $mc_j(q)$ for q residents. Each atomistic renter must choose one unit of housing and has preferences over amenities and consumption, and we can describe their utility using the value function $u_{i,j} = x_j - r_j$. In the competitive spatial equilibrium, (1) all renters are housed in their utility maximizing location, (2) utility is equal everywhere, and (3) and zero profits. This conditions imply that (1) $\sum_j q_j = N$, (2) $u_{i,j} = \bar{u}$, and (3) $mc_j(q_j) = r_j$.

To find a solution, we note that (2) implies that $r_j = x_j - \bar{u}$, that (3) implies $q_j = mc_j^{-1}(r_j)$, and together with (1) we have $\sum_j \left(mc_j^{-1}(x_j - \bar{u})\right) = N$. These equations form a system of $J+1$ unknowns ($\{\vec{r}, \bar{u}\}$), $J+1$ parameters ($\{\vec{x}, N\}$), and $J+1$ equations:

$$\sum_j \left(mc_j^{-1}(x_j - \bar{u})\right) = N$$
$$r_j = x_j - \bar{u}.$$

The above equations make clear that changes in marginal cost at j do not impact rents r_j. Furthermore, were the outside option $\bar{u}$ to be endogenously determined through N, say through idiosyncratic tastes for the outside option, the impact of a cost shock at j on rent r_j would operate only through the outside option $\bar{u}$, i.e. insofar as marginal costs at j contribute to a market-level cost shock. Thus, if landlords are atomistic, then an idiosyncratic cost shock will not affect $\bar{u}$ nor rent.

A.8.2 Markups in the Monocentric City Model

In this section, we adapt the monocentric city model to include individual heterogeneity in location preferences for a finite set of locations that induces downward sloping residual demand for locations. This heterogeneity implies that markups exist and that profits are increasing with proximity to the city center. In addition, prices in denser areas are higher and the quantity of housing lower than if the supply were competitive (price set to marginal cost). Pricing power results in a city that is flatter, demonstrating how pricing power acts as a dispersive force.

We begin with the model from Appendix A.8.1, which immediately precedes this section, noting that x_j can denote location j's distance to the city center on a line segment, with differences between any two locations j and $j+1$'s distances x_j and x_{j+1} being constant and nonzero for all js (such that there are a large but finite set of locations J), starting with the CBD itself at $j = 0$ with distance $x_0 = 0$, until an endogenously determined periphery point,$j^\flat$ with distance to CBD $x_{j^\flat}$.[39] In addition we assume a mass N of renters and for simplicity we assume that the city is closed. The assumption of a discrete number of locations follows Berliant and Fujita (1992) and is a departure from standard monocentric city models, where space is continuous for tractability reasons, but similar to Alonso's original model.

[39]Note, defining x_J as the furthest point from the CBD that is feasible given the landscape (possibly due to natural features), we allow for $x_{j^\flat} \leq x_J$.

We now adjust the above model by positing that indirect utility of each individual i at location j is a combination of location j's rent, r_j, and distance to the CBD, x_j, as well as an idiosyncratic valuation parameter, ϵ_i, drawn from some distribution $G(\epsilon)$ where $\epsilon \geq 0$ and $\mathsf{E}[\epsilon] \in \mathcal{R}_+$:

$$v_i(j) = (1 - x_j)\epsilon_i - r_j. \tag{63}$$

Note that this is isomorphic to heterogeneous transportation costs.[40]

Downward-sloping demand for each location is derived from renter heterogeneity and the assertion that for each distance x_j there is only 1 plot of that distance (with differences $x_j - x_{j+1} > 0$ and constant). An alternative approach that would sustain markups in the presence of multiple plots with identical values x_j would be to assume renter-location specific heterogeneity $\epsilon_{i,j}$ as in the main text discrete choice model. In such a setting, markups can converge to a positive constant even as the number of choices goes to infinity.

In the present setting, willingness to pay for each location will be different for each individual. This generates a downward sloping residual demand curve for each location j that brings about markups. As in the previous model, willingness to pay is also determined by prices at other locations:

$$WTP_i(j) = \min_{j'}\{(1 - x_j)\epsilon_i - (1 - x_{j'})\epsilon_i - r_{j'}\}. \tag{64}$$

Note that for any given option j', the difference between i's valuations of j and j' is increasing in ϵ_i. This super-modularity guarantees a sorting equilibrium: each location j will be populated by a set of types $[\underline{\epsilon}_j, \bar{\epsilon}_j]$, with higher locations having higher cutoffs (Mirrlees, 1971). Locally, the only relevant alternatives for individuals at plot j are at plots $j - 1$ and $j + 1$. Specifically, at each plot j there is a threshold, $\tilde{\epsilon}_j$, below which the relevant alternative is $j + 1$, and above which the relevant alternative is $j - 1$: the highest types in each plot will be deciding between j and moving one space closer to the city center while the lower types will be deciding between j and moving space further.

Locally, conditional on $\epsilon_i > \tilde{\epsilon}_j$, willingness to pay for j is therefore:

$$WTP_i(j) = (x_{j-1} - x_j)\epsilon_i - r_{j-1}, \tag{65}$$

[40]In addition, we assume that all units are the constrained to be equal sized.

while below $\tilde{\epsilon}_j$, willingness to pay for j is:

$$WTP_i(j) = (x_{j+1} - x_j)\epsilon_i - r_{j+1}. \tag{66}$$

Total demand for j given a price r_j is therefore

$$q_j(r_j) = P(\epsilon > \tilde{\epsilon}_j) \cdot P\left(\epsilon < \frac{r_{j-1} - r_j}{x_j - x_{j-1}}\right) + P(\epsilon \le \tilde{\epsilon}_j) \cdot P\left(\epsilon > \frac{r_j - r_{j+1}}{x_{j+1} - x_j}\right) \tag{67}$$

or

$$q_j(r_j) = [1 - G(\tilde{\epsilon}_j)] \cdot G\left(\frac{r_{j-1} - r_j}{x_j - x_{j-1}}\right) + G(\tilde{\epsilon}_j) \cdot \left[1 - G\left(\frac{r_j - r_{j+1}}{x_{j+1} - x_j}\right)\right]. \tag{68}$$

It's clear from the above that quantity demanded is declining in price on two margins: as r_j increases, those at the top margin near $\bar{\epsilon}_j$ leave for $j-1$ and those at the bottom margin $\underline{\epsilon}_j$ leave for $j+1$. Marginal revenue is therefore also declining. As in the main text, profits are given by:

$$\pi = \max_{\mathbb{1}_{redev}, q_j} \begin{cases} r_j(q_j) \cdot q_j - C^{\mathsf{d}}(q_j) - C^{\mathsf{f}}(q_j) & \text{if } \mathbb{1}_{redev} = 1 \\ 0 & \text{if } \mathbb{1}_{redev} = 0, \end{cases} \tag{69}$$

where here we assume for simplicity no structure currently exists at any location j, no transfers, and costs are symmetric. We also now assert a fixed cost of development $C^{\mathsf{d}}(0) > 0$.

At each location, taking prices at all other plots $r_{j'}$ as given, landlords set quantities to maximize profits and will do so where falling marginal revenues are equal to marginal cost. Because prices must be decreasing with distance x_j in the sorting equilibrium described above, r_{j+1} and r_{j-1} are also falling with x_j. All else equal, demand at any price is falling in x_j. It follows that plots closer to the city will have higher quantities and profits. It further follows that the city will be filled in: if a location j is developed, all locations $j' < j$ will also be developed.

To close the model, the peripheral location, j^{b} at distance $x_{j^{\mathsf{b}}}$, must be found after which no development occurs. Note that quantity demanded at j^{b} is given by:

$$q^D_{j^{\mathsf{b}}}(r_{j^{\mathsf{b}}}) = G\left(\frac{r_{j^{\mathsf{b}}-1} - r_{j^{\mathsf{b}}}}{x_{j^{\mathsf{b}}} - x_{j^{\mathsf{b}}-1}}\right). \tag{70}$$

A slight complication to note is that, as is present in the above demand expression, at this location (as well as at the most central location) there is only a single margin from which to draw additional individuals, or to which marginal individuals leave.

Because space is discrete, we have an approximation of a zero profit condition in Eq 71. The peripheral location, $j^{\mathbf{b}}$, is defined as the furthest plot from the city center such that its profit is weakly positive (and almost zero) but at the next undeveloped plot, $(j^{\mathbf{b}}+1)$, profit upon entry is strictly negative:

$$0 > r_{(j^{\mathbf{b}}+1)}(q_{(j^{\mathbf{b}}+1)}) \cdot q_{(j^{\mathbf{b}}+1)} - C^{\mathsf{d}}(q_{(j^{\mathbf{b}}+1)}) - C^{\mathsf{f}}(q_{(j^{\mathbf{b}}+1)}). \tag{71}$$

Despite (near) zero profits, the landlord at $j^{\mathbf{b}}$ still sets rent with a markup over marginal cost. Rather, the fixed cost $C^{\mathsf{d}}(0)$ almost exactly offsets variable profit from the optimally chosen quantity at the periphery (and profits are increasing towards the city center).

Finally, the total quantity of housing provided must equal the total quantity demanded:

$$\mathsf{N} = \sum_{0}^{j^{\mathbf{b}}} q_j^*(r_j^*), \tag{72}$$

where q_j^* and r_j^* are equilibrium, profit-maximizing quantities and prices at plot j.

To see monopoly power act as a dispersion force, note that at the city center, $j = 0$, demand is given by:

$$q_0^D(r_0) = 1 - G\left(\frac{r_0 - r_1}{x_1}\right). \tag{73}$$

Quantity at this location is lower than in a competitive model due to both landlord at $j = 0$ setting rent above marginal cost *and*, through general equilibrium forces: rent at $j = 1$ is also set its marginal cost. However, not every location will experience reduced density: the equilibrium effects of reduced density at the center push demand out to the periphery, and, at some point, these effects may dominate the pricing power effect and locations further away may be more built up than in under marginal cost pricing. Furthermore, the increased profitability at $j^{\mathbf{b}}$ and the general equilibrium effects increasing demand at $j^{\mathbf{b}}$ imply that the peripherally developed plot is *futher* away from the CBD under monopolistic pricing. Taken together, these facts imply the density of the monopolistically priced city first-order stochastically dominates that of the city built to marginal cost pricing (although we do not formally show this).

We reiterate that the markup in this setting is a feature of two items: heterogeneity of individual taste in the form of the (variance of the) distribution $G(\epsilon)$, and amenity differences $x_j - x_{j-1}$. As either go to zero, markups are eliminated. This is **not** the case if instead heterogeneity took the form of individual-by-location draws, $\epsilon_{i,j}$, as in the model in the main text. Under this regime with specific assumptions on the distribution of $G(\epsilon)$, markups will converge to a constant as the distance between spaces converges to zero or, as on a disk, multiple buildings with the same value of x coexist.

A.8.3 Entry and Exit to Other Real Markets

Our baseline model takes the number of parcels available for rent as constant. An alternative, and more realistic, assumption is that landlords can choose which of several housing markets to enter. For example, a landlord with an *ex ante* rental building could sell it as condominiums or a commercial building could be converted to residential space.

In such a setup, the city overall is still comprised of an exogenous set of parcels, $\mathcal{A} = \{a_0, a_1, a_2, \ldots, a_J\}$, but each housing market, m, consists of an endogenously selected and mutually exclusive subset of parcels, $\bigcup_m \mathcal{A}_m = \mathcal{A}$, for all real estate markets m. Once these sets are determined equations (1)-(9) remain the same as in the main text. To close the model and determine these sets, we add an addition equilibrium constraint. Facing different demand (and potentially costs) in each market, each landlord chooses which housing market to enter, such that the chosen market is the profit-maximizing market for that parcel:

$$m_a^* = \underset{m}{\operatorname{argmax}}\{\pi_a^m\} \tag{74}$$

Exit and entry change the plots's profitability in their incumbent markets. In the case of symmetric costs and demand for all buildings in each market, this yields profit equalization condition across markets.

Because of the real costs of these kinds of switches and policy-determined constraints, we observe relatively few buildings switching between markets over the course of our sample. Still such a setup might be more descriptive of a very long-run land-use equilibrium.

A.8.4 Generalized Free Entry

Another alternative assumption is that there is a fringe of parcels with the capacity to freely enter the market. That is, undeveloped land that could be turned into apartment buildings. This arrangement is difficult to fathom in a built urban core, such as our empirical setting in New York, but could be appropriate for locations proximate to large undeveloped terrain.

In this case, again, the set of parcels available for renters is endogenous in the model, and—again—conditional on this decision, equations (1)-(9) remain unchanged. An easy structure to close such a model is to assume a fixed entry cost ξ_e (e.g., a permitting cost), before a draw of the location amenity a. Although we do not specify profitability as a function of a, the equilibrium condition determining the set $\mathcal{A}$ is that profits are zero in expectation:

$$\mathsf{E}[\pi_a] = \xi_e \tag{75}$$

In this case, expected profits are zero, but realized profits may be heterogeneous. Ex-post of entry, landlords still charge markups. While profits are zero, in general, efficiency of this market structure depends on more restrictive assumptions on preferences (Bilbiie, Ghironi, and Melitz, 2008).

It should also be noted that under this market structure, typical analyses of urban policies are highly non-regular. For example, the welfare impact of zoning can become ambiguous, as parcel-level deviations from efficiency could be ameliorated by the policy's (potential) effect of induced entry. For this reason, and because the assumption of free entry is unrealistic in the urban core, we do not favor this analysis.

A.8.5 Uncertainty

We consider uncertainty in the form of uncertain (future) building-specific demand, so that the joint developer-landlord expected profit function is:

$$\mathsf{E}[\pi_a] = \max_{\mathbb{1}_{redev}, q_a} \begin{cases} \mathsf{E}\left[\mu \cdot r_a(q_a) \cdot q_a - C_a^{\mathsf{d}}(q_a) - C_a^{\mathsf{f}}(q_a) + \mathsf{S}_a\right] & \text{if } \mathbb{1}_{redev} = 1 \\ \mathsf{E}\left[\mu \cdot r_a(q_a) \cdot q_a - C_a^{\mathsf{f}}(q_a)\right] & \text{if } \mathbb{1}_{redev} = 0 \end{cases} \tag{76}$$

where μ is an exogenous, stochastic parameter capturing the uncertainty of future demand with distribution $G(\mu)$ that is non-negative and has finite mean and variance. In this case, the developer-landlord's optimal choice of q (conditional on the redevelopment) and the redevelopment decision only depend on $E(\mu)$, rather than the realized value. Because the demand system is stochastic, the social surplus is also stochastic, and a social planner optimizes over the same expected value, $E(\mu)$. Redevelopment failure occurs, if it occurs, for the non-stochastic demand system $E[\mu] \cdot r_a(q_a)$, and Proposition 1 remains unchanged.

This approach differs from Titman (1985) in that our redevelopment decision must be decided upon before the realized state of the world. We also differ from Capozza and Li (1994), who model uncertainty in a continuous time framework with Brownian motion.

The latter framework is fitting for a separate research question: *what dictates the timing of redevelopment decisions?* Though beyond the scope of this paper to analyse, we note that as is standard in the investment literature, demand variance generates option value and postpones investment. As in He and Kondor (2016), sub-optimal redevelopment may occur.

A.8.6 Endogenous Quality Selection

Here we consider a model extension where developers are allowed the additional choice of adjusting building quality when considering redevelopment.

Buildings, indexed by j, can be in one of a set of discrete, hierarchical quality categories $a \in A = \{a_1, a_2, ...a_Q\}$, where we say $a_{j,0}$ is the initial quality.[41] In addition, each building has some initial capacity, $q_{j,0}$. We let the as-is sale price of the building vary across buildings, even conditional on initial quality and size, due to various factors such as different capital vintages and depreciation. Developer d, who owns a parcel containing a building with attributes $(a_{j,0}, q_{j,0})$, chooses whether to sell the building as-is or to redevelop, and, conditional on redeveloping, chooses a quality, a, from the discrete set of potential quality types and a new quantity, q, conditional on long-run zoning regulations.

Thus, the developer's new maximization problem is:

$$\pi^{\mathsf{d}} = \max_{\{\mathbb{1}_{redev}, a, q\}} \begin{cases} s_j(a_{j,0}, q_{j,0}) & \text{if } \mathbb{1}_{redev} = 0 \\ [s(a,q) - C(q,a)] \cdot \xi_a^{\mathsf{d}} & \text{if } \mathbb{1}_{redev} = 1 \end{cases} \qquad (77)$$

$$\text{s.t. } q \leq q_{j,z},$$

where we now assume the cost function and sale prices are quality but not developer specific, and the value of redevelopment to a specific quality a has an idiosyncratic component ξ_a^{d}, which is developer and quality specific, drawn from a Fréchet distribution with shape parameter θ: $F(\xi) = \mathrm{e}^{\xi^{-\theta}}$.

Solving the developer problem in reverse may be instructive. Because all buildings of a specific quality face same cost function and residual demand, all developers choosing to redevelop to a given quality will also choose to redevelop to the same quantity, which we refer to as q_a.[42] Embedding the quantity choice into the quality choice, the problem now

[41] Note, we no longer index buildings by a, as in the main text, since this is now a choice variable.

[42] For a given developer conditional on a choice of a, choosing q over $q' \neq q$ implies that $s(a,q) - C(a,q) > s(a,q') - C(a,q')$. Thus, as long as there is a unique optimal quantity conditional on the quality choice, then

becomes a standard discrete choice problem with Fréchet draws. In particular, *conditional on redevelopment*, the probability of choosing quality a is:

$$\Pr\left(a \mid \mathbb{1}_{redev} = 1\right) = \frac{(s(a, q_a) - C(a, q_a))^{\theta}}{\sum_{a' \in A}(s(a', q_{a'}) - C(a', q_{a'}))^{\theta}}. \tag{78}$$

Furthermore, conditional on a non-redeveloped building sale price of $\tilde{s}_0$, which maps to a set of initial building qualities and sizes, the share of buildings being redeveloped to quality level (a, q_a) is:

$$\int_{\tilde{s}_0} \Pr(\pi_a > \tilde{s}_0) = \int_{\tilde{s}_0} \mathsf{e}^{-(s(a,q_a)-C(a,q_a))^{-\theta}} \mathsf{d}G(\tilde{s}), \tag{79}$$

where $G(\cdot)$ is the distribution on sales prices conditional on non-development.

Therefore, the number of redeveloped parcels in each period and to each quality level can be found by summing the above expression over the distribution of building valuations across the city.

A number of interesting new dynamics build off of the results of the model in our main text. For example, through markups, exogenous changes in demand for luxury housing may lead to switches from low income to luxury redevelopment and increase the pricing power of lower income housing landlords.

all developers choosing a given a, will choose the same q.

B HHI and Ownership Matching

Here we describe how we match buildings to owner groups. This procedure is necessary because a large portion of reported rental building owners are corporate entities.[43] Thus the reported ownership structure underestimates the degree of common ownership. The NYC Department of Housing Preservation and Development (HPD) requires that building owners register each building with multiple dwellings (or that is inhabited primarily by non-family members) and compiles this registration list to create the Multiple Dwelling Registry and Contacts (MDRC). Importantly, the MDRC assigns a unique ID to each building-owner pair and for each owner lists the names of the main 'shareholders' of the corporate owner. Building owners must re-register annually. Thus, we have a list of buildings with their corporate owner names and a list of shareholder names linked to corporate owners.

However, we face two data challenges in matching buildings to owners using the MDRC.[44] First, we only have MDRC lists for scattered years: 2012, 2014, 2015, 2017, 2018, 2020, and 2022. The MDRC lists are updated quarterly by the HPD; however, vintage files are not stored publicly. Thus, we have collected these files based on archived web-pages and, for 2012 and 2015, through the generosity of the NYU Furman Center. Second, the MDRC does not link buildings by common owners and is not built to do this; rather, the files are designed only to identify a single building's owner. Thus, there is no identifier that links owners across buildings other than their names. We deal with each in turn.

To create a building owner panel, we append the MDRC annual files together and 'back-fill' the ownership from MDRC information for missing years. That is, if we observe a building-owner pair for year 2020, then we assume the owner is the same from 2020, 2019, 2018, and so on, until we have another observation. When constructing the MDRC, we arrange the shareholder names based on frequency.[45]

We then merge this with our DOF/PLUTO building year panel of rental buildings. Table A1 reports the match rate for the main four boroughs by year used in the rent sample.

Finally, we use a text matching procedure to ensure that the reported building corporate

[43]We use the term 'corporate entity' loosely; some building owners are truly corporations while others are limited liability companies, sole proprietorships, partnerships, or cooperatives.

[44]Technically, the MDRC are two different files: the Multiple Dwelling List (MDL) and the Registration Contacts (RD) list. We merge the two together by a common building identifier.

[45]For example, if name A is associated with 5 buildings and name B with 4 buildings, then for any set of buildings with both names $\{A, B\}$ we designate name A as the primary name.

Table A1: Match Rate Across Boroughs

	BX	BK	MN	QN
2007	0.94	0.93	0.96	0.89
2008	0.94	0.92	0.96	0.89
2009	0.94	0.92	0.95	0.88
2010	0.94	0.92	0.96	0.89
2011	0.94	0.92	0.96	0.89
2012	0.94	0.92	0.96	0.88
2013	0.94	0.92	0.96	0.88
2014	0.94	0.92	0.96	0.88
2015	0.94	0.92	0.96	0.88
2016	0.94	0.92	0.96	0.88
2017	0.94	0.92	0.96	0.88
2018	0.94	0.92	0.96	0.88
2019	0.93	0.91	0.96	0.88

Note: 2007-2019 NYC residential buildings with 4+ units. Data from DOF, PLUTO, MDRC files. Match rate between from PLUTO-FAR and MDRC lists.

owner matches the MDRC corporate owner name. We find that, even though we match over 90% of our buildings to the MDRC, we only match the reported owners about 60% of the time. We can treat these buildings two different ways. First, we could ignore the difference in owners between the two datasets, which could exaggerate concentration. Second, we could treat these non-matches as unique owners, which could attenuate concentration. We pursue the latter in the interest of providing the most conservative results.

To find all buildings that have common shareholders, we again perform a text matching procedure across buildings within a Census tract-year pair. We perform this procedure for each tract-year pair in the four main boroughs of NYC.[46] Our procedure calculates multiple five scores: one for the amount of matching whole words ('tokens') and then four for sequences of characters ('n-grams').[47] We assign multiple threshold value rules to these scores to determine whether there is a match. For example, if two owners have over 70% of the same 4-grams in the same order, then we call this a match. For another

[46] As a robustness test, we also use the NYC geography Neighborhood Tabulation Areas (NTAs) that are collections of Census tracts within Public Use Micro Areas (PUMAs). We hesitate to expand beyond these areas as our matching procedure uses only first and last names; e.g., there could be many owners named Joe Smith across the city but the chance is lower within a local geography.

[47] We calculate 3-grams, 4-grams, 8-grams, and 10-grams.

example, if two owners have over 5% of 10-grams and at least four matching tokens, then this is a match. We tested simpler threshold rules (e.g., 'if over 50% of 3-grams match, then match'), but we were not satisfied when manually auditing the results. For any building that does not match to the MDRC, we use the reported ownername (usually a corporate entity) and require an exact string match within the tract-year.